

Article

New Two-Parameter Ridge Estimators for Addressing Multicollinearity in Linear Regression: Theory, Simulation, and Applications

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Abstract

Multicollinearity among explanatory variables often undermines the reliability of the ordinary least squares (OLS) estimator that can be used in linear regression modeling. To overcome the limitation, a variety of two-parameter estimation strategies have been developed in prior research. We revisit these existing methods and present a newly established two-parameter ridge estimator to improve the accuracy of regression coefficients in terms of multicollinearity settings. A theoretical evaluation, assessed under the mean squared error (MSE) framework, is examined to compare their efficiency. Furthermore, a comprehensive simulation study is conducted to examine the empirical properties of all these estimators for different configurations, followed by a real-life dataset to examine their performance.

Keywords: ordinary least squares (OLS); multicollinearity; mean squared error (MSE); new two-parameter estimator; Monte Carlo simulation; real data application

1. Introduction

Traditional linear regression analysis commonly relies on the assumption that the predictor variables are independent of one another. When the assumption is fulfilled, the ordinary least squares estimator achieves optimal statistical performance and satisfies the conditions of the best linear unbiased estimator (BLUE). In practical data analysis, predictor variables frequently exhibit moderate to strong correlations, violating the independence assumption and producing the multicollinearity problem. Under the multicollinearity problem, the explanatory (predictor) variables share redundant information, and the OLS estimator becomes inefficient; its estimated coefficients may have inflated standard errors, incorrect signs, and wide confidence intervals (Nayem et al., 2024) [1].

Multicollinearity is therefore regarded as a key challenge that affects regression modeling. Various approaches have been proposed to mitigate it, including: (i) expanding the sample size, (ii) eliminating predictors with a high correlation, (iii) combining variables through linear transformations, (iv) using dimension reduction via principal component analysis (PCA), (v) utilizing partial least squares regression (PLSR), and (vi) applying biased estimation techniques like ridge regression and related extensions. Among these, ridge regression, which was originally proposed by Hoerl and Kennard (1970 [2]), is used as the most widely influential approach.

Since its inception, many alternative biased estimators have been developed to improve the OLS estimator because of the presence of multicollinearity (Hoerl and Kennard,



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1970 [2]). The classical ridge regression estimator introduces a positive scalar, k , known as the ridge or biasing parameter, that stabilizes estimates but whose performance depends critically on its proper selection. Later, Liu (1993 [3]) proposed an alternative biased estimator that introduces a different biasing parameter, d , which enters linearly and offers computational simplicity and interpretational advantages.

To further enhance estimation, several two-parameter estimators have been proposed that combine both k and d and provide greater flexibility and potential efficiency gains compared to single-parameter estimators such as ridge or Liu regression. Several influential studies in this area include the works of Lukman et al. (2019 [4]), Yang and Chang (2010 [5]), and Owolabi et al. (2022 [6]), among others. Although different two-parameter estimators have been proposed to address multicollinearity, several limitations remain. So, these existing two-parameter estimators rely on chosen shrinkage parameters, which may not adequately adapt to the eigen structure of the design matrix. Also, theoretical comparison results are often only derived under restrictive conditions, with limited empirical validation across a broad range of correlation levels and error variances. Finally, practical guidance on when newly proposed estimators outperform classical ridge or Liu-type estimators is often lacking in the literature.

The study undertakes a comparative evaluation of the numerous established two-parameter estimators and also presents a newly developed two-parameter ridge estimator that extends our classical ridge framework through an eigenvalue-adaptive biasing structure. We present theoretical comparisons using the MSE criterion and conduct an extensive Monte Carlo simulation across multiple scenarios. Additionally, we demonstrate their practical applicability by using empirical data from a real-life data application.

The proposed estimator involves two tuning parameters that play distinct but complementary statistical roles. The first parameter primarily controls the overall degree of shrinkage applied to the regression coefficients, thereby regulating the bias–variance trade-off in the presence of multicollinearity. Larger values of this parameter induce stronger shrinkage, which reduces the variance inflation caused by highly correlated predictors at the expense of a controlled increase in bias.

The second parameter governs the direction and structure of the shrinkage by incorporating information from the eigen structure of the design matrix. In particular, this parameter allows for differential shrinkage along principal component directions associated with large eigenvalues of $X^T X$, enabling the estimator to adapt to the underlying correlation pattern among regressors. As a result, the estimator stabilizes coefficient estimates in ill-conditioned regression settings more effectively than one-parameter ridge-type estimators.

The rest of this manuscript is organized in the following manner. Section 2 introduces the statistical materials and methods and the full set of the estimators and in Section 3, their theoretical results are compared under the MSE criterion. In Section 4, we discuss the results of the Monte Carlo simulation design and also present the numerical findings and provide a practical application using a real-world dataset. And, in Section 5, we conclude the paper with key findings and possible future extensions.

2. Materials and Methods

This section explores the linear regression framework and discusses various types of estimators developed for the linear regression model.

2.1. Regression and Estimator Construction

Consider the linear regression model of the following form:

$$Y = X\beta + \epsilon, \epsilon \sim N(0, \sigma^2 I_n), \quad (1)$$

where Y denotes an $n \times 1$ response vector variable, X represents an $n \times p$ design matrix of regressors with full rank column matrix, β is a $q \times 1$ vector of unknown regression coefficients, and the random error ϵ is assumed an $n \times 1$ vector of residuals, which is assumed to be normally distributed with $E(\epsilon) = 0$, and $Var(\epsilon) = \sigma^2 I_n$, where I_n is the $n \times n$ identity matrix.

The ordinary least square estimator of β in Equation (1) is defined as follows:

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T Y, \quad (2)$$

where X is the design matrix.

Let R be an orthogonal matrix, which is $R^T X^T X R = \Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p)$, where λ_j is the j th eigen value of $X^T X$. Λ and R represent the diagonal matrix of eigen values and the corresponding matrix of eigen vectors of $X^T X$, respectively. Using the transformation, Equation (1) can be equivalently written as follows:

$$Y = Z\alpha + \epsilon, \quad (3)$$

where $Z = XR$, $\alpha = R^T \beta$ and $Z^T Z = \Lambda$.

Now, the OLS estimator of α under the transformed model is therefore as follows:

$$\hat{\alpha}_{OLS} = \Lambda^{-1} Z^T Y \quad (4)$$

The statistical properties of the OLS estimator are

$$\text{Bias}(\hat{\alpha}_{OLS}) = 0, \quad (5)$$

$$\text{Cov}(\hat{\alpha}_{OLS}) = \sigma^2 \Lambda^{-1}, \quad (6)$$

where $\hat{\sigma}^2 = \frac{(Y - Z\hat{\alpha}_{OLS})^T (Y - Z\hat{\alpha}_{OLS})}{n-p-1}$

$$\begin{aligned} \text{MMSE}(\hat{\alpha}_{OLS}) &= \text{Cov}(\hat{\alpha}_{OLS}) + \text{Bias}(\hat{\alpha}_{OLS})\text{Bias}(\hat{\alpha}_{OLS})^T \\ &= \sigma^2 \Lambda^{-1}, \end{aligned} \quad (7)$$

$$\begin{aligned} \text{MSE}(\hat{\alpha}_{OLS}) &= E[(\hat{\alpha}_{OLS} - \alpha)^T (\hat{\alpha}_{OLS} - \alpha)] \\ &= \text{tr}(\text{MMSE}(\hat{\alpha}_{OLS})) = \sigma^2 \sum_{j=1}^p \frac{1}{\lambda_j} \end{aligned} \quad (8)$$

2.2. Two-Parameter Estimators

This section examines different types of two-parameter estimators that have previously been introduced in different studies.

2.2.1. Liu Type Two-Parameter Estimator

Liu (2003) [7] has proposed a two-parameter estimator, which is designed to mitigate multicollinearity and is defined as

$$\hat{\alpha}_{LTE} = (\Lambda + kI)^{-1} (Z^T Y - d\alpha^*), \quad (9)$$

where α^* is any estimator of α , and we choose $\alpha^* = \alpha_{OLS}$ and then we can get

$$\hat{\alpha}_{LTE} = (\Lambda + kI)^{-1} (\Lambda - dI) \hat{\alpha}_{OLS}.$$

The Bias, the covariance matrix, MMSE and scalar MSE are given, respectively:

$$\text{Bias}(\hat{\alpha}_{LTE}) = -(d+k)(\Lambda + kI)^{-1}\alpha,$$

$$\text{Cov}(\hat{\alpha}_{LTE}) = \sigma^2 A_{LTE} \Lambda^{-1} A_{LTE}^T,$$

where $A_{LTE} = (\Lambda + kI)^{-1}(\Lambda - dI)$.

$$\text{MMSE}(\hat{\alpha}_{LTE}) = \sigma^2 A_{LTE} \Lambda^{-1} A_{LTE}^T + (d+k)^2 (\Lambda + kI)^{-1} \alpha \alpha^T (\Lambda + kI)^{-1} \quad (10)$$

$$\text{MSE}(\hat{\alpha}_{LTE}) = \sigma^2 \sum_{j=1}^p \frac{(d-\lambda_j)^2}{\lambda_j(\lambda_j+k)^2} + \sum_{j=1}^p \frac{(d+k)^2 \alpha_j^2}{(\lambda_j+k)^2} \quad (11)$$

For choosing k and d in this case from Liu (2003) [7],

$$\hat{k}_{LTE} = \frac{\lambda_1 - 100 * \lambda_p}{99}, \quad \hat{d}_{LTE} = \frac{\sum_{j=1}^p \left(\frac{(\hat{\sigma}^2 - k\alpha_j^2)}{(\lambda_j + \hat{k})^2} \right)}{\sum_{j=1}^p \left(\frac{(\hat{\sigma}^2 + \lambda_j \alpha_j^2)}{\lambda_j(\lambda_j + \hat{k})^2} \right)}$$

2.2.2. Ozkale and Kaciranlar Two-Parameter Estimator

An additional two-parameter estimator, developed by Ozkale and Kaciranlar (2007), [8] is expressed as follows:

$$\hat{\alpha}_{TP} = A_{TP} \hat{\alpha}_{OLS}, \quad (12)$$

where $A_{TP} = (\Lambda + kI)^{-1}(\Lambda + kdI)$.

Here, Bias, the covariance matrix, MMSE and scalar MSE of $\hat{\alpha}_{TP}$ are as follows:

$$\text{Bias}(\hat{\alpha}_{TP}) = k(d-1)(\Lambda + kI)^{-1}\alpha,$$

$$\text{Cov}(\hat{\alpha}_{TP}) = \sigma^2 A_{TP} \Lambda^{-1} A_{TP}^T,$$

$$\text{MMSE}(\hat{\alpha}_{TP}) = \sigma^2 A_{TP} \Lambda^{-1} A_{TP}^T + k^2(d-1)^2 (\Lambda + kI)^{-1} \alpha \alpha^T (\Lambda + kI)^{-1}, \quad (13)$$

$$\text{MSE}(\hat{\alpha}_{TP}) = \sigma^2 \sum_{j=1}^p \frac{(\lambda_j + kd)^2}{\lambda_j(\lambda_j+k)^2} + \sum_{j=1}^p \frac{k^2(d-1)^2 \alpha_j^2}{(\lambda_j+k)^2}. \quad (14)$$

Now, using the approach of Ozkale and Kaciranlar (2007) [8], the optimal of d for the fixed k value can be derived as follows:

$$\hat{d}_{opt} = \frac{\sum_{j=1}^p \frac{(k\alpha_j^2 - \hat{\sigma}^2)}{(\lambda_j + k)^2}}{\sum_{j=1}^p \frac{k(\hat{\sigma}^2 + \lambda_j \alpha_j^2)}{\lambda_j(\lambda_j + k)^2}},$$

and

$$\hat{k} = \frac{\hat{\sigma}^2}{\hat{\alpha}_j^2 - d \left(\frac{\hat{\sigma}^2}{\lambda_j} + \alpha_j^2 \right)}$$

Hoerl et al. (1975) [9] suggested estimating k by using the harmonic mean values of k , which was originally proposed by Hoerl and Kennard (1970) [2]. Consequently, both harmonic and arithmetic averaging can be used to estimate this shrinkage parameter. Then,

$$\hat{k}_{AM} = \frac{1}{p} \sum_{j=1}^p \frac{\hat{\sigma}^2}{\hat{\alpha}_j^2 - d \left(\frac{\hat{\sigma}^2}{\lambda_j} + \alpha_j^2 \right)} \text{ and } \hat{k}_{HM} = \frac{p\hat{\sigma}^2}{\sum_{j=1}^p \left[\hat{\alpha}_j^2 - d \left(\frac{\hat{\sigma}^2}{\lambda_j} + \alpha_j^2 \right) \right]}.$$

Since the $\hat{d}_{opt}$ depends on the k and the estimation of k also depends on the d , an iterative strategy is required for these parameters.

2.2.3. New Biased Estimator Based on Ridge

Proposed by Sakallioğlu and Kiciranlar (2008) [10], this is defined as

$$\begin{aligned} \hat{\alpha}_{NBE} &= (\Lambda + I)^{-1} (Z^T y + d \hat{\alpha}_{Ridge}) \\ &= (\Lambda + I)^{-1} (\Lambda + (d + k)I) (\Lambda + kI)^{-1} \Lambda \hat{\alpha}_{OLS}. \end{aligned} \quad (15)$$

The Bias, the covariance matrix, MMSE and scalar MSE are given, respectively,

$$Bias(\hat{\alpha}_{NBE}) = (A_{NBE} - I)\alpha,$$

$$Cov(\hat{\alpha}_{NBE}) = \sigma^2 A_{NBE} \Lambda^{-1} A_{NBE}^T,$$

where $A_{NBE} = (\Lambda + I)^{-1} (\Lambda + (d + k)I) (\Lambda + kI)^{-1} \Lambda$.

$$MMSE(\hat{\alpha}_{NBE}) = \sigma^2 A_{NBE} \Lambda^{-1} A_{NBE}^T + (A_{NBE} - I) \alpha \alpha^T (A_{NBE} - I)^T, \quad (16)$$

$$MSE(\hat{\alpha}_{NBE}) = \sigma^2 \sum_{j=1}^p \frac{\lambda_j (\lambda_j + (d + k))^2}{(\lambda_j + 1)^2 (\lambda_j + k)^2} + \sum_{j=1}^p \frac{((1 - d)\lambda_j + k)^2 \alpha_j^2}{(\lambda_j + k)^2 (\lambda_j + 1)^2}. \quad (17)$$

To estimate the unknown parameter k and d , one may select the following:

$$\hat{d}_{opt} = \frac{\sum_{j=1}^p \frac{\lambda_j (\hat{\alpha}_j^2 - \hat{\sigma}^2)}{(\lambda_j + 1)^2 (\lambda_j + k)}}{\sum_{j=1}^p \frac{\lambda_j (\lambda_j \hat{\alpha}_j^2 + \hat{\sigma}^2)}{(\lambda_j + 1)^2 (\lambda_j + k)^2}}.$$

$$\text{For fixed } k, k_{HK} = \frac{\hat{\sigma}^2}{\sum_{j=1}^p \hat{\alpha}_j^2} \text{ and } k_{HKB} = \frac{p\hat{\sigma}^2}{\sum_{j=1}^p \hat{\alpha}_j^2}.$$

2.2.4. Yang and Chang Two-Parameter Estimator

A two-parameter estimator, which is proposed by Yang and Chang (2010) [5], is defined as follows:

$$\hat{\alpha}_{YC} = Y_{k,d} \hat{\alpha}_{OLS}, \quad (18)$$

where $Y_{k,d} = (\Lambda + I)^{-1} (\Lambda + dI) (\Lambda + kI)^{-1} \Lambda$, and k and d are biasing parameters.

The Bias, the covariance matrix, MMSE and scalar MSE of $\hat{\alpha}_{YC}$ are as follows:

$$\begin{aligned} Bias(\hat{\alpha}_{YC}) &= (Y_{k,d} - I)\alpha, \\ Cov(\hat{\alpha}_{YC}) &= \sigma^2 Y_{k,d} \Lambda^{-1} Y_{k,d}^T, \end{aligned}$$

$$MMSE(\hat{\alpha}_{YC}) = \sigma^2 Y_{k,d} \Lambda^{-1} Y_{k,d}^T + (Y_{k,d} - I) \alpha \alpha^T (Y_{k,d} - I)^T, \quad (19)$$

$$MSE(\hat{\alpha}_{YC}) = \sigma^2 \sum_{j=1}^p \frac{\lambda_j (\lambda_j + d)^2}{(\lambda_j + 1)^2 (\lambda_j + k)^2} + \sum_{j=1}^p \frac{((k + 1 - d)\lambda_j + k)^2 \alpha_j^2}{(\lambda_j + k)^2 (\lambda_j + 1)^2}. \quad (20)$$

For k , we fixed d ; we get the optimal k as

$$\hat{k}_j = \frac{\hat{\sigma}^2(\lambda_j + d) - (1 - d)\lambda_j\hat{\alpha}_j^2}{(\lambda_j + 1)\hat{\alpha}_j^2}$$

We can apply different approaches for estimating the parameters, such as the arithmetic mean and median.

We get

$$\hat{d}_{opt} = \frac{\sum_{i=1}^p \left[((k+1)\lambda_j + k)\lambda_j\hat{\alpha}_j^2 - \lambda_j^2\hat{\sigma}^2 \right] / \left[(\lambda_j + 1)^2(\lambda_j + k)^2 \right]}{\sum_{i=1}^p \left(\hat{\sigma}^2 + \lambda_j\hat{\alpha}_j^2 \right) \lambda_j / \left[(\lambda_j + 1)^2(\lambda_j + k)^2 \right]}.$$

2.2.5. Modified Almost Unbiased Liu Estimator

Proposed by Arumairajan and Wijekoon (2017) [11], this is defined as follows:

$$\hat{\alpha}_{MAULE} = A_{MAULE}\hat{\alpha}_{OLS}, \quad (21)$$

where $A_{MAULE} = (I - (1 - d)^2(\Lambda + I)^{-2})(\Lambda + kI)^{-1}\Lambda$.

The Bias, the covariance matrix, MMSE and scalar MSE of $\hat{\alpha}_{MAULE}$ are as follows:

$$\begin{aligned} \text{Bias}(\hat{\alpha}_{MAULE}) &= (A_{MAULE} - I)\alpha, \\ \text{Cov}(\hat{\alpha}_{MAULE}) &= \sigma^2 A_{MAULE} \Lambda^{-1} A_{MAULE}^T, \\ \text{MMSE}(\hat{\alpha}_{MAULE}) &= \sigma^2 A_{MAULE} \Lambda^{-1} A_{MAULE}^T + (A_{MAULE} - I)\alpha\alpha^T(A_{MAULE} - I)^T, \quad (22) \\ \text{MSE}(\hat{\alpha}_{MAULE}) &= \sigma^2 \sum_{j=1}^p \frac{\lambda_j(\lambda_j + 2 - d)^2(\lambda_j + d)^2}{(\lambda_j + k)^2(\lambda_j + 1)^4} \\ &\quad + \sum_{j=1}^p \frac{[\lambda_j(\lambda_j + 2 - d)(\lambda_j + d) - (\lambda_j + k)(\lambda_j + 1)^2]^2 \alpha_j^2}{(\lambda_j + k)^2(\lambda_j + 1)^4}. \end{aligned} \quad (23)$$

$$\text{Now, } \hat{d}_{opt} = 1 - \sqrt{\frac{\sum_{j=1}^p \frac{\lambda_j(\sigma^2 - k\alpha_j^2)}{(\lambda_j + k)^2(\lambda_j + 1)^2}}{\sum_{j=1}^p \frac{\lambda_j(\sigma^2 + \lambda_j\alpha_j^2)}{(\lambda_j + 1)^4(\lambda_j + k)^2}}}, \text{ and } \hat{k} = \frac{\sigma^2(\lambda_j + 1)^2\lambda_j - (1 - d)^2\lambda_j(\sigma^2 + \alpha_j^2)}{\alpha_j^2(\lambda_j + 1)^2}.$$

We can obtain the arithmetic mean value of k .

2.2.6. Modified Almost Unbiased Two-Parameter Estimator

Proposed by Lukman, Adewuyi, Oladejo, and Olukayode (2019) [4], this is defined as follows:

$$\hat{\alpha}_{MAUTP} = A_{MAUTP}\hat{\alpha}_{OLS}, \quad (24)$$

where $A_{MAUTP} = (I - k^2(1 - d)^2(\Lambda + kI)^{-2})(\Lambda + kI)^{-1}\Lambda$.

The Bias, the covariance matrix, MMSE and scalar MSE of $\hat{\alpha}_{MAUTP}$ are as follows:

$$\begin{aligned} \text{Bias}(\hat{\alpha}_{MAUTP}) &= (A_{MAUTP} - I)\alpha, \\ \text{Cov}(\hat{\alpha}_{MAUTP}) &= \sigma^2 A_{MAUTP} \Lambda^{-1} A_{MAUTP}^T, \\ \text{MMSE}(\hat{\alpha}_{MAUTP}) &= \sigma^2 A_{MAUTP} \Lambda^{-1} A_{MAUTP}^T + (A_{MAUTP} - I)\alpha\alpha^T(A_{MAUTP} - I)^T, \quad (25) \end{aligned}$$

$$MSE(\hat{\alpha}_{MAUTP}) = \sigma^2 \sum_{j=1}^p \frac{\lambda_j (\lambda_j + 2k - kd)^2 (\lambda_j + kd)^2}{(\lambda_j + k)^3} + \sum_{j=1}^p \frac{[\lambda_j (\lambda_j + 2k - kd) (\lambda_j + kd) - (\lambda_j + k)^3]^2 \alpha_j^2}{(\lambda_j + k)^2}. \quad (26)$$

To estimate k and d , we use $\hat{k} = \frac{\sigma^2}{\alpha_j^2}$ and the harmonic version of $\hat{k}$ is $\hat{k}_{HMP} = \frac{p\hat{\sigma}^2}{\sum_{j=1}^p \hat{\alpha}_j^2}$ and $\hat{d} = \min \left(\frac{\alpha_j^2}{\frac{\sigma^2}{\alpha_j^2} + \alpha_j^2} \right)$.

2.2.7. Modified Ridge Type

A modified ridge-type estimator (MRT) is proposed by Lukman et al. (2019) [12], which is defined as follows:

$$\hat{\alpha}_{MRT} = M_{k,d} \hat{\alpha}_{OLS}, \quad (27)$$

where $M_{k,d} = (\Lambda + k(1+d)I)^{-1} \Lambda$, k and d are biasing parameters.

The Bias, the covariance matrix, MMSE and scalar MSE of $\hat{\alpha}_{MRT}$ are as follows:

$$Bias(\alpha_{MRT}) = -k(1+d) \left((\Lambda + k(1+d)I)^{-1} \right) \alpha,$$

$$Cov(\hat{\alpha}_{MRT}) = \sigma^2 M_{k,d} \Lambda^{-1} M_{k,d}^T,$$

$$MMSE(\hat{\alpha}_{MRT}) = \sigma^2 M_{k,d} \Lambda^{-1} M_{k,d}^T + k^2 (1+d)^2 (\Lambda + k(1+d)I)^{-1} \alpha \alpha^T (\Lambda + k(1+d)I)^{-1}, \quad (28)$$

$$MSE(\hat{\alpha}_{MRT}) = \sigma^2 \sum_{j=1}^p \frac{\lambda_j}{(\lambda_j + k(1+d))^2} + \sum_{j=1}^p \frac{k^2 (1+d)^2 \alpha_j^2}{(\lambda_j + k(1+d))^2}. \quad (29)$$

Following Lukman et al. (2019) [12]:

$$\hat{k}_j = \frac{\hat{\sigma}^2}{(1+d)\hat{\alpha}_j^2}.$$

The harmonic mean of the $\hat{k}_j$ is

$$\hat{k}_{HMP} = \frac{p\hat{\sigma}^2}{\sum_{j=1}^p (1+d)\hat{\alpha}_j^2},$$

and

$$\hat{d}_j = \frac{\hat{\sigma}^2}{k\hat{\alpha}_j^2} - 1.$$

Also, the harmonic means of $\hat{d}_j$ is as follows:

$$\hat{d}_{MRT} = \frac{p}{\sum_{j=1}^p \left(\frac{1}{\hat{d}} \right)}.$$

2.2.8. A New Biased Estimator

A new biased two-parameter estimator was developed by Dawoud and Kibria (2020) [13] and is defined as follows:

$$\hat{\alpha}_{DK} = A_{DK} \hat{\alpha}_{OLS}, \quad (30)$$

where $A_{DK} = (\Lambda + k(1+d)I)^{-1} (\Lambda - k(1+d)I)$.

The Bias, the covariance matrix, MMSE and scalar MSE of $\hat{\alpha}_{DK}$ are as follows:

$$\text{Bias}(\hat{\alpha}_{DK}) = -2k(1+d)\alpha(\Lambda + k(1+d)I)^{-1},$$

$$\text{Cov}(\hat{\alpha}_{DK}) = \sigma^2 A_{DK} \Lambda^{-1} A_{DK}^T,$$

$$\text{MMSE}(\hat{\alpha}_{DK}) = \sigma^2 A_{DK} \Lambda^{-1} A_{DK}^T + 4k^2(1+d)^2(\Lambda + k(1+d)I)^{-1} \alpha \alpha^T (\Lambda + k(1+d)I)^{-1}, \quad (31)$$

$$\text{MSE}(\hat{\alpha}_{DK}) = \sigma^2 \sum_{j=1}^p \frac{(\lambda_j - k(1+d))^2}{\lambda_j(\lambda_j + k(1+d))^2} + \sum_{j=1}^p \frac{4k^2(1+d)^2 \alpha_j^2}{(\lambda_j + k(1+d))^2}. \quad (32)$$

For k , we fixed d ; we get the optimal k as

$$\hat{k}_j = \frac{\hat{\sigma}^2}{(1+\hat{d})\left(\frac{\hat{\sigma}^2}{\lambda_j} + 2\hat{\alpha}_j^2\right)},$$

and $k_{\min} = \min(\hat{k}_j)$.

Then, we get

$$\hat{d}_{opt} = \frac{\hat{\sigma}^2 \lambda_j}{m} - 1,$$

where $m = \hat{k}(\hat{\sigma}^2 + 2\lambda_j \hat{\alpha}_j^2)$.

2.2.9. Generalized Two-Parameter Estimator

Proposed by Zeinal (2020) [14], this is defined as follows:

$$\hat{\alpha}_{GTP} = (\Lambda + kI)^{-1}(\Lambda + kD)\hat{\alpha}_{OLS}, \quad (33)$$

where $D = \text{diag}(d_1, d_2, \dots, d_p)$.

The Bias, the covariance matrix, MMSE and scalar MSE of $\hat{\alpha}_{GTP}$ are as follows:

$$\text{Bias}(\hat{\alpha}_{GTP}) = (A_{GTP} - I)\alpha,$$

$$\text{Cov}(\hat{\alpha}_{GTP}) = \sigma^2 A_{GTP} \Lambda^{-1} A_{GTP}^T.$$

$$\text{MMSE}(\hat{\alpha}_{GTP}) = \sigma^2 A_{GTP} \Lambda^{-1} A_{GTP}^T + (A_{GTP} - I) \alpha \alpha^T (A_{GTP} - I)^T, \quad (34)$$

$$\text{MSE}(\hat{\alpha}_{GTP}) = \sigma^2 \sum_{j=1}^p \frac{(\lambda_j + kd_j)^2}{\lambda_j(\lambda_j + k)^2} + \sum_{j=1}^p \frac{k^2(d_j - 1)^2 \alpha_j^2}{(\lambda_j + k)^2}, \quad (35)$$

where $A_{GTP} = (\Lambda + kI)^{-1}(\Lambda + kD)$.

For the unknown parameter, we get the optimal $d_j, j = 1, 2, \dots, p$ for fixed k

$$\hat{d}_{jopt} = \frac{(k\hat{\alpha}_j^2 - \hat{\sigma}^2)\lambda_j}{k(\hat{\sigma}^2 + \hat{\alpha}_j^2 \lambda_j)}, \quad j = 1, 2, \dots, p$$

then, we get

$$\hat{k}_j = \frac{\hat{\sigma}^2}{\hat{\alpha}_j^2 - d_j\left(\frac{\hat{\sigma}^2}{\lambda_j} + \hat{\alpha}_j^2\right)},$$

and the arithmetic mean of the above k_j .

2.2.10. Modified Liu Ridge Type

Proposed by Aslam and Ahmad (2022) [15], this is defined as

$$\hat{\alpha}_{MLRT} = A_{MLRT} \hat{\alpha}_{OLS}, \quad (36)$$

where $A_{MLRT} = (\Lambda + I)^{-1}(\Lambda + dI)(\Lambda + k(1 + d)I)^{-1}\Lambda$.

The Bias, the covariance matrix, MMSE and scalar MSE of $\hat{\alpha}_{MLRT}$ are as follows:

$$Bias(\hat{\alpha}_{MLRT}) = (A_{MLRT} - I)\alpha,$$

$$Cov(\hat{\alpha}_{MLRT}) = \sigma^2 A_{MLRT} \Lambda^{-1} A_{MLRT}^T,$$

$$MMSE(\hat{\alpha}_{MLRT}) = \sigma^2 A_{MLRT} \Lambda^{-1} A_{MLRT}^T + (A_{MLRT} - I)\alpha\alpha^T(A_{MLRT} - I)^T, \quad (37)$$

$$MSE(\hat{\alpha}_{MLRT}) = \sigma^2 \sum_{j=1}^p \frac{\lambda_j(\lambda_j + d)^2}{(\lambda_j + k(1 + d))^2(\lambda_j + 1)^2} + \sum_{j=1}^p \frac{(\lambda_j(1 - d) + k(1 + d)(\lambda_j + 1))^2 \alpha_j^2}{(\lambda_j + k(1 + d))^2(\lambda_j + 1)^2}. \quad (38)$$

$$\text{Now, } k_{opt} = \frac{\sigma^2(\lambda_j + d) - \lambda_j(1 - d)\alpha_j^2}{(1 + d)(\lambda_j + 1)\alpha_j^2}, \text{ and } d_{opt} = \frac{\sum_{j=1}^p \frac{(\lambda_j + k(\lambda_j + 1))[\lambda_j^2 - k\lambda_j^2 - k\lambda_j]\alpha_j^2 - \sigma^2\lambda_j[\lambda_j^2 - k\lambda_j^2 - k\lambda_j]}{(\lambda_j + 1)^2}}{\sum_{j=1}^p \frac{\sigma^2[\lambda_j^2 - k\lambda_j^2 - k\lambda_j] + (\lambda_j - k(\lambda_j + 1))[\lambda_j^2 - k\lambda_j^2 - k\lambda_j]\alpha_j^2}{(\lambda_j + 1)^2}}.$$

2.2.11. New Biased Regression Two-Parameter Estimator

Dawoud, Lukman and Haadi (2022) [16] proposed the following estimator,

$$\hat{\alpha}_{NBR} = RWM\hat{\alpha}_{OLS}, \quad (39)$$

where $R = (\Lambda + kI)^{-1}(\Lambda + kdI)$ and WM for KL and $W = (\Lambda + kI)^{-1}$ and $M = (\Lambda - kI)$.

The Bias, the covariance matrix, MMSE and scalar MSE matrix of $\hat{\alpha}_{NBR}$ are as follows:

$$Bias(\hat{\alpha}_{NBR}) = (RWM - I)\alpha,$$

$$Cov(\hat{\alpha}_{NBR}) = \sigma^2 RWM\Lambda^{-1}M^T W^T R^T,$$

$$MMSE(\hat{\alpha}_{NBR}) = \sigma^2 RWM\Lambda^{-1}M^T W^T R^T + (RWM - I)\alpha\alpha^T(RWM - I)^T, \quad (40)$$

$$MSE(\hat{\alpha}_{NBR}) = \sigma^2 \sum_{j=1}^p \frac{(\lambda_j - k)^2(\lambda_j + kd)^2}{\lambda_j(\lambda_j + k)^4} + \sum_{j=1}^p \frac{[(\lambda_j + kd)(\lambda_j - k) - (\lambda_j + k)]^2 \alpha_j^2}{(\lambda_j + k)^4}. \quad (41)$$

$$\text{Here, } \hat{k} = \frac{-(\lambda_j^2 \alpha_j^2 (3 - d) + \sigma^2 \lambda_j (1 - d))}{2(\sigma^2 d + \lambda_j \alpha_j^2 (1 + d))} + \frac{\lambda_j \sqrt{\lambda_j (\alpha_j^2)^2 (d - 3)^2 + 2\lambda_j \sigma^2 \alpha_j^2 (5 - 2d + d^2) + (\sigma^2)^2 (1 + d)^2}}{2(\sigma^2 d + \lambda_j \alpha_j^2 (1 + d))}, \text{ and}$$

we take the minimum values.

$$\text{And } \hat{d} = \frac{\lambda_j^2 (\sigma^2 - 3\alpha_j^2 k) - \lambda_j k (\sigma^2 + \alpha_j^2 k)}{k(k - \lambda_j)(\sigma^2 + \lambda_j \alpha_j^2)} \text{ and the minimum values of this.}$$

2.2.12. Biased Two-Parameter Estimator

Idowu, Oladapo, Owolabi and Ayinde (2022) [17] proposed this biased two-parameter estimator and it is defined as follows:

$$\hat{\alpha}_{BTP} = EH\hat{\alpha}_{OLS}, \quad (42)$$

where $E = (\Lambda + I)^{-1}(\Lambda - dI)$ and $H = (\Lambda + k(1 + d)I)^{-1}(\Lambda - k(1 + d)I)$.

The Bias, the covariance matrix, MMSE and scalar MSE of $\hat{\alpha}_{BTP}$ are as follows:

$$\begin{aligned} \text{Bias}(\hat{\alpha}_{BTP}) &= (EH - I)\alpha, \\ \text{Cov}(\hat{\alpha}_{BTP}) &= \sigma^2 EH\Lambda^{-1}H^TE^T, \\ \text{MMSE}(\hat{\alpha}_{BTP}) &= \sigma^2 EH\Lambda^{-1}H^TE^T + (EH - I)\alpha\alpha^T(EH - I)^T, \end{aligned} \quad (43)$$

$$\begin{aligned} \text{MSE}(\hat{\alpha}_{BTP}) &= \sigma^2 \sum_{j=1}^p \frac{(\lambda_j - d)^2 (\lambda_j - k(1+d))^2}{\lambda_j (\lambda_j + 1)^2 (\lambda_j + k(1+d))^2} \\ &+ \sum_{j=1}^p \frac{[(\lambda_j - d)(\lambda_j - k(1+d)) - (\lambda_j + 1)(\lambda_j + k(1+d))]^2 \alpha_j^2}{(\lambda_j + 1)^2 (\lambda_j + k(1+d))^2}. \end{aligned} \quad (44)$$

$$\text{Now, } \hat{k} = \frac{\sigma^2 \lambda_j (d - \lambda_j) + \alpha_j^2 \lambda_j^2 (d + 1)}{\sigma^2 (1 + d)(d - \lambda_j) - \alpha_j^2 \lambda_j (1 + d)(2\lambda_j - d + 1)}, \text{ and,}$$

$$\hat{d} = \frac{-(\sigma^2 \lambda_j - \sigma^2 k + \alpha_j^2 \lambda_j^2 + \sigma^2 \lambda_j k + 2\alpha_j^2 \lambda_j^2 k)}{2(\sigma^2 k + \alpha_j^2 \lambda_j k)} + \frac{\sqrt{(\alpha_j^2)^2 \lambda_j^2 k(2\lambda_j k + k + \lambda_j) + \sigma^2 \alpha_j^2 \lambda_j^2 k(k - \lambda_j) + \sigma^2 \alpha_j^2 \lambda_j k(2\sigma^2 \lambda_j k + k + \lambda_j) + (\sigma^2)^2 \lambda_j k(k - \lambda_j)}}{(\sigma^2 k + \alpha_j^2 \lambda_j k)}.$$

2.2.13. New Two-Parameter Estimator

A new two-parameter estimator has been developed by Owolabi, Ayinde, Idowu, Oladapo and Lukman (2022) [18]:

$$\hat{\alpha}_{NTP} = A_{NTP} \hat{\alpha}_{OLS}, \quad (45)$$

where $A_{NTP} = (\Lambda + kdI)^{-1}(\Lambda - kdI)$.

The Bias, the covariance matrix, MMSE and scalar MSE matrix of $\hat{\alpha}_{NTP}$ are as follows:

$$\begin{aligned} \text{Bias}(\hat{\alpha}_{NTP}) &= (A_{NTP} - I)\alpha, \\ \text{Cov}(\hat{\alpha}_{NTP}) &= \sigma^2 A_{NTP} \Lambda^{-1} A_{NTP}^T, \\ \text{MMSE}(\hat{\alpha}_{NTP}) &= \sigma^2 A_{NTP} \Lambda^{-1} A_{NTP}^T + (A_{NTP} - I)\alpha\alpha^T(A_{NTP} - I)^T, \end{aligned} \quad (46)$$

$$\text{MSE}(\hat{\alpha}_{NTP}) = \sigma^2 \sum_{j=1}^p \frac{(\lambda_j - kd)^2}{\lambda_j (\lambda_j + kd)^2} + \sum_{j=1}^p \frac{4k^2 d^2 \alpha_j^2}{(\lambda_j + kd)^2}. \quad (47)$$

$$\text{Now, } \hat{k} = \frac{\sigma^2}{d(2\alpha_j^2 + \frac{\sigma^2}{\lambda_j})} \text{ and the harmonic mean is } \hat{k}_{HM} = \frac{\sigma^2}{\Sigma d(2\alpha_j^2 + \frac{\sigma^2}{\lambda_j})}.$$

$$\text{And } \hat{d}_j = \frac{\sigma^2}{k(2\alpha_j^2 + \frac{\sigma^2}{\lambda_j})}.$$

2.2.14. New Ridge-Type Estimator

The new ridge-type two-parameter estimator proposed by Owolabi, Ayinde, and Alabi (2022) [6] is defined as follows:

$$\hat{\alpha}_{NRT} = A_{NRT} \hat{\alpha}_{OLS}, \quad (48)$$

where $A_{NRT} = (\Lambda + (k + d)I)^{-1}\Lambda$.

The Bias, the covariance matrix, MMSE and scalar MSE matrix of $\hat{\alpha}_{NRT}$ are as follows:

$$\begin{aligned} \text{Bias}(\hat{\alpha}_{NRT}) &= (A_{NRT} - I)\alpha, \\ \text{Cov}(\hat{\alpha}_{NRT}) &= \sigma^2 A_{NRT} \Lambda^{-1} A_{NRT}^T, \end{aligned}$$

$$MMSE(\hat{\alpha}_{NRT}) = \sigma^2 A_{NRT} \Lambda^{-1} A_{NRT}^T + (A_{NTP} - I) \alpha \alpha^T (A_{NTP} - I)^T, \quad (49)$$

$$MSE(\hat{\alpha}_{NRT}) = \sigma^2 \sum_{j=1}^p \frac{\lambda_j}{(\lambda_j + (k+d))^2} + \sum_{j=1}^p \frac{(k+d)^2 \alpha_j^2}{(\lambda_j + (k+d))^2}. \quad (50)$$

$$\text{Now, } \hat{k} = \min\left(\frac{\sigma^2}{\alpha_j^2} - d\right) \text{ and } \hat{d} = \min\left(\frac{\sigma^2}{\alpha_j^2} - k\right).$$

2.2.15. Liu–Kibria–Lukman Two-Parameter Estimator

Proposed by Idowu et al. (2023) [19], the two-parameter estimator is defined as follows:

$$\hat{\alpha}_{LKL} = CA\hat{\alpha}_{OLS}, \quad (51)$$

where $C = (\Lambda + I)^{-1}(\Lambda + dI)$ and $A = (\Lambda + kI)^{-1}(\Lambda - kI)$.

The Bias, the covariance matrix, MMSE and scalar MSE of $\hat{\alpha}_{LKL}$ are as follows:

$$\text{Bias}(\hat{\alpha}_{LKL}) = (CA - I)\alpha,$$

$$\text{Cov}(\hat{\alpha}_{LKL}) = \sigma^2 CA\Lambda^{-1}A^T C^T,$$

$$MMSE(\hat{\alpha}_{LKL}) = \sigma^2 CA\Lambda^{-1}A^T C^T + (CA - I)\alpha\alpha^T(CA - I)^T, \quad (52)$$

$$MSE(\hat{\alpha}_{LKL}) = \sigma^2 \sum_{j=1}^p \frac{(\lambda_j - k)^2 (\lambda_j + d)^2}{\lambda_j (\lambda_j + k)^2 (\lambda_j + 1)^2} + \sum_{j=1}^p \frac{[(\lambda_j + d)(\lambda_j - k) - (\lambda_j + k)(\lambda_j + 1)]^2 \alpha_j^2}{(\lambda_j + k)^2 (\lambda_j + 1)^2}. \quad (53)$$

$$\text{Now, } \hat{d} = \frac{\lambda_j (\alpha_j^2 - \sigma^2) + \lambda_j k (2\alpha_j^2 \lambda_j + \alpha_j^2 - \sigma^2)}{\sigma^2 (\lambda_j - k) + \alpha_j^2 \lambda_j (\lambda_j - k)}.$$

For the biasing parameter k , the proposed LKL estimator is given as follows:

$$k = \min \left[\frac{\sigma^2}{2\alpha_j^2 + \frac{\sigma^2}{\lambda_j}} \right].$$

2.2.16. Two-Parameter Ridge Estimator

Proposed by Lipovetsky and Conklin (2005) [20], a two-parameter ridge estimator is defined as

$$\hat{\alpha}_{TPR} = q(\Lambda + kI)^{-1} \Lambda \hat{\alpha}_{OLS}, \quad (54)$$

$$\text{where } q = \frac{(Z^T Y)^T (\Lambda + kI)^{-1} Z^T Y}{(Z^T Y)^T (\Lambda + kI)^{-1} \Lambda (\Lambda + kI)^{-1} Z^T Y}.$$

The Bias, the covariance matrix, MMSE and scalar MSE of $\hat{\alpha}_{TPR}$ are as follows:

$$\text{Bias}(\hat{\alpha}_{TPR}) = (A_{TPR} - I)\alpha,$$

$$\text{Cov}(\hat{\alpha}_{TPR}) = \sigma^2 A_{TPR} \Lambda^{-1} A_{TPR}^T,$$

where $A_{TPR} = q(\Lambda + kI)^{-1} \Lambda$.

$$MMSE(\hat{\alpha}_{TPR}) = \sigma^2 A_{TPR} \Lambda^{-1} A_{TPR}^T + (A_{TPR} - I)\alpha\alpha^T(A_{TPR} - I)^T, \quad (55)$$

$$MSE(\hat{\alpha}_{TPR}) = \sigma^2 \sum_{j=1}^p \frac{q^2 \lambda_j}{(\lambda_j + k)^2} + \sum_{j=1}^p \frac{(q\lambda_j - \lambda_j - k)^2 \alpha_j^2}{(\lambda_j + k)^2}. \quad (56)$$

Now, we use the value of $k = \left(\prod_{j=1}^p \left(\frac{\lambda_j}{|\alpha_j|} \right)^2 \right) \left(\frac{\sigma^2}{\alpha_{max}^2} \right)$ by Khan et al. (2024) [21]

and this $\hat{k}$ is utilized by Toker and Kaçiranlar (2013) [22] from this $\hat{q} = \frac{\sum_{j=1}^p \frac{\alpha_j^2 \lambda_j}{\lambda_j + k}}{\sum_{j=1}^p \frac{\sigma^2 \lambda_j + \alpha_j^2 \lambda_j^2}{(\lambda_j + k)^2}}$ for

computing the initial corresponding value of $\hat{q}$.

The values of $\hat{q}$ can be obtained by the corresponding values of $\hat{k}_{opt}$.

$$\hat{k}_{opt} = \frac{q \sum_{j=1}^p \sigma^2 \lambda_j + (q-1) \sum_{j=1}^p \alpha_j^2 \lambda_j^2}{\sum_{j=1}^p \alpha_j^2 \lambda_j^2}.$$

2.2.17. New Proposed Minimum Eigenvalue Two-Parameter Ridge Estimator

Previous two-parameter ridge estimator introduced by Lipovetsky and Conklin (2005) [20], which is defined as

$$\hat{\alpha}_{TPR} = q(\Lambda + kI)^{-1} \Lambda \hat{\alpha}_{OLS},$$

where

$$q = \frac{(Z^T Y)^T (\Lambda + kI)^{-1} Z^T Y}{(Z^T Y)^T (\Lambda + kI)^{-1} \Lambda (\Lambda + kI)^{-1} Z^T Y},$$

which can be written as

$$q = \frac{(Z^T Y)^T (\Lambda + kI)^{-1} (\Lambda + kI) (\Lambda + kI)^{-1} Z^T Y}{(Z^T Y)^T (\Lambda + kI)^{-1} \Lambda (\Lambda + kI)^{-1} Z^T Y}.$$

But the ridge regression obtained by Hoerl and Kennard (1970) [2] is $\hat{\alpha}(k) = (\Lambda + kI)^{-1} Z^T Y$. Then, q can be defined as

$$q = \frac{\hat{\alpha}^T(k) (\Lambda + kI) \hat{\alpha}(k)}{\hat{\alpha}^T(k) \Lambda \hat{\alpha}(k)}$$

Then using the min-max principle, it can be shown that the value of q will be between

$$l_1 \leq q \leq l_p$$

where l_1 and l_p are the minimum and maximum eigen value of $[\Lambda + kI](\Lambda)^{-1}$. From a preliminary simulation study, we found that the two-parameter estimator performed better when $q = l_1$. Therefore, we propose the following new ridge regression estimator,

$$\hat{\alpha}_{NTPR} = l_1 (\Lambda + kI)^{-1} \Lambda \hat{\alpha}_{OLS}, \quad (57)$$

where l_1 is the minimum eigen value of the matrix $[\Lambda + kI](\Lambda)^{-1}$.

Now, the Bias, the covariance matrix, MMSE and scalar MSE of $\hat{\alpha}_{NTPR}$ are as follows:

$$Bias(\hat{\alpha}_{NTPR}) = (A_{NTPR} - I)\alpha,$$

$$Cov(\hat{\alpha}_{NTPR}) = \sigma^2 A_{NTPR} \Lambda^{-1} A_{NTPR}^T,$$

where $A_{NTPR} = l_1 (\Lambda + kI)^{-1} \Lambda$.

$$MMSE(\hat{\alpha}_{TPR}) = \sigma^2 A_{NTPR} \Lambda^{-1} A_{NTPR}^T + (A_{NTPR} - I) \alpha \alpha^T (A_{NTPR} - I)^T, \quad (58)$$

$$MSE(\hat{\alpha}_{NTPR}) = \sigma^2 \sum_{j=1}^q \frac{l_1^2 \lambda_j}{(\lambda_j + k)^2} + \sum_{j=1}^q \frac{(l_1 \lambda_j - \lambda_j - k)^2 \alpha_j^2}{(\lambda_j + k)^2}. \quad (59)$$

Now, we use the value of $k = \left(\sum_{j=1}^p \left(\frac{\lambda_j}{|\alpha_j|} \right)^2 \right) \left(\frac{\sigma^2}{\alpha_{max}^2} \right)$ by Khan et al. (2024) [21]

and this $\hat{k}$ is utilized by Tokar and Kaçiranlar (2013) [22] from this $\hat{l}_1 = \frac{\sum_{j=1}^p \frac{\alpha_j^2 \lambda_j}{\lambda_j + k}}{\sum_{j=1}^p \frac{\sigma^2 \lambda_j + \alpha_j^2 \lambda_j^2}{(\lambda_j + k)^2}}$ for

computing the initial corresponding value of $\hat{l}_1$.

The values of $\hat{l}_1$ are used to compute the corresponding optimum values of $\hat{k}_{opt}$.

$$\hat{k}_{opt} = \frac{l_1 \sum_{j=1}^p \sigma^2 \lambda_j + (l_1 - 1) \sum_{j=1}^p \alpha_j^2 \lambda_j^2}{\sum_{j=1}^p \alpha_j^2 \lambda_j^2}.$$

3. Results

In this section, we will discuss the results of the theoretical comparison for our new proposed estimator. The notations and lemmas given below are very useful for comparative analysis between these estimators.

Lemma 1. Let $n \times n$ matrices $M > 0$, $N > 0$ ($N \geq 0$), then $M > N$ if and only if $\lambda_j(NM^{-1}) < 1$, where $\lambda_j(NM^{-1})$ is the largest eigen value of matrix NM^{-1} .

Lemma 2. (Farebrother, 1976 [23]): Let M be an $n \times n$ positive definite matrix, let α be a non-zero $n \times 1$ column matrix and let c be a positive scalar. Then, $cM - \alpha\alpha^T > 0$ if $\alpha^T M^{-1} \alpha < d$, and $cM - \alpha\alpha^T \geq 0$ if $\alpha^T M^{-1} \alpha \leq d$.

Lemma 3. (Trenkler and Toutenberg, 1990 [24]): Let $\hat{\alpha}_j = A_j y$, $j = 1, 2$ be two linear estimators of α . Suppose that $D = \text{Cov}(\hat{\alpha}_1) - \text{Cov}(\hat{\alpha}_2) > 0$, where $\text{Cov}(\hat{\alpha}_j)$, $j = 1, 2$ denotes the covariance matrix of $\hat{\alpha}_j$ and $b_j = \text{Bias}(\hat{\alpha}_j) = (A_j X - I) \alpha$, $j = 1, 2$. Consequently,

$$\Delta(\hat{\alpha}_1 - \hat{\alpha}_2) = \text{MMSE}(\hat{\alpha}_1) - \text{MMSE}(\hat{\alpha}_2) = \sigma^2 D + b_1 b_1^T - b_2 b_2^T > 0.$$

If and only if $b_2^T [\sigma^2 D + b_1 b_1^T] b_2 < 1$, where $\text{MMSE}(\hat{\alpha}_j) = \text{Cov}(\hat{\alpha}_j) + b_j b_j^T$.

3.1. Theoretical Comparison of Some Estimators

This section presents a theoretical comparison of the estimators, based on the methodologies proposed by Hoque and Kibria (2023) [25]. While there are many possible methods to be compared, we are presenting a few of the comparisons in this section and rest can be compared in a similar way.

3.1.1. Comparison Between New Proposed Two-Parameter Ridge Estimator and OLS Estimator

The difference between $\text{MMSE}(\hat{\alpha}_{NTPR})$ and $\text{MMSE}(\hat{\alpha}_{OLS})$ can be obtained by

$$\text{MMSE}(\hat{\alpha}_{OLS}) - \text{MMSE}(\hat{\alpha}_{NTPR}) = \sigma^2 \Lambda^{-1} - \sigma^2 A_{NTPR} \Lambda^{-1} A_{NTPR}^T - b_{NTPR} b_{NTPR}^T. \quad (60)$$

If we let $k > 0$, we have the following theorem.

Theorem 1. If $k > 0$, $b_{NTPR} = \text{Bias}(\hat{\alpha}_{NTPR})$ is the bias of NTPR, we can say that $\hat{\alpha}_{NTPR}$ is better than the estimator $\hat{\alpha}_{OLS}$ using the theorem of MMSE, that is, $\text{MMSE}(\hat{\alpha}_{OLS}) - \text{MMSE}(\hat{\alpha}_{NTPR}) > 0$, if and only if

$$b_{NTPR}^T \left[\sigma^2 \left(\Lambda^{-1} - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) \right]^{-1} b_{NTPR} < 1.$$

Proof. Difference between the scalar MSE functions,

$$\begin{aligned} \text{MMSE}(\hat{\alpha}_{OLS}) - \text{MMSE}(\hat{\alpha}_{NTPR}) &= \sigma^2 \left(\Lambda^{-1} - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) - b_{NTPR} b_{NTPR}^T \\ &= \sigma^2 \text{diag} \left\{ \frac{1}{\lambda_j} - \frac{l_1^2 \lambda_j}{(\lambda_j + k)^2} \right\}_{j=1}^p - b_{NTPR} b_{NTPR}^T, \end{aligned} \quad (61)$$

where $\Lambda^{-1} - A_{NTPR} \Lambda^{-1} A_{NTPR}^T$ is positive definite and we know that it is possible if and only if $(\lambda_j + k)^2 - l_1^2 \lambda_j > 0$. For $k > 0$, we get that $k^2 + 2\lambda_j k + \lambda_j^2 (1 - l_1^2) > 0$, and we can conclude that $\Lambda^{-1} - A_{NTPR} \Lambda^{-1} A_{NTPR}^T$ is a positive definite. Using lemma 2, the proof is completed. $\square$

3.1.2. Comparison Between New Proposed Two-Parameter Ridge Estimator and Liu Type Two-Parameter Estimator

The difference between $\text{MMSE}(\hat{\alpha}_{NTPR})$ and $\text{MMSE}(\hat{\alpha}_{LTE})$ is obtained by

$$\text{MMSE}(\hat{\alpha}_{LTE}) - \text{MMSE}(\hat{\alpha}_{NTPR}) = \sigma^2 A_{LTE} \Lambda^{-1} A_{LTE}^T - \sigma^2 A_{NTPR} \Lambda^{-1} A_{NTPR}^T + b_{LTE} b_{LTE}^T - b_{NTPR} b_{NTPR}^T. \quad (62)$$

If we let $k > 0$, we have the following theorem.

Theorem 2. If $k > 0$ and $0 < d < 1$, and $b_{LTE} = \text{Bias}(\hat{\alpha}_{LTE})$ is the bias of LTE, we can say that the $\hat{\alpha}_{NTPR}$ is better than the estimator $\hat{\alpha}_{LTE}$ using the theorem of MMSE, that is, $\text{MMSE}(\hat{\alpha}_{LTE}) - \text{MMSE}(\hat{\alpha}_{NTPR}) > 0$, if and only if

$$b_{NTPR}^T \left[\sigma^2 \left(A_{LTE} \Lambda^{-1} A_{LTE}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) + b_{LTE} b_{LTE}^T \right]^{-1} b_{NTPR} < 1.$$

Proof. Difference between scalar MSE functions,

$$\begin{aligned} \text{MMSE}(\hat{\alpha}_{LTE}) - \text{MMSE}(\hat{\alpha}_{NTPR}) &= \sigma^2 \left(A_{LTE} \Lambda^{-1} A_{LTE}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) + b_{LTE} b_{LTE}^T - b_{NTPR} b_{NTPR}^T \\ &= \sigma^2 \text{diag} \left\{ \frac{(d - \lambda_j)^2}{\lambda_j (\lambda_j + k)^2} - \frac{l_1^2 \lambda_j}{(\lambda_j + k)^2} \right\}_{j=1}^p + b_{LTE} b_{LTE}^T - b_{NTPR} b_{NTPR}^T, \end{aligned} \quad (63)$$

where $A_{LTE} \Lambda^{-1} A_{LTE}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T$ is positive definite and we know that it is possible if and only if $(d - \lambda_j)^2 - l_1^2 \lambda_j^2 > 0$. For $k > 0$ and $0 < d < 1$, we get that $d^2 - 2\lambda_j + \lambda_j^2 (1 - l_1^2) > 0$, and we can conclude that $A_{LTE} \Lambda^{-1} A_{LTE}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T$ is a positive definite. By lemma 2, the proof is completed. $\square$

3.1.3. Comparison Between New Proposed Two-Parameter Ridge Estimator and Ozkale and Kaciranlar Two-Parameter Estimator

The difference between $\text{MMSE}(\hat{\alpha}_{NTPR})$ and $\text{MMSE}(\hat{\alpha}_{TP})$ is obtained by

$$\text{MMSE}(\hat{\alpha}_{TP}) - \text{MMSE}(\hat{\alpha}_{NTPR}) = \sigma^2 A_{TP} \Lambda^{-1} A_{TP}^T - \sigma^2 A_{NTPR} \Lambda^{-1} A_{NTPR}^T + b_{TP} b_{TP}^T - b_{NTPR} b_{NTPR}^T. \quad (64)$$

If we let $k > 0$, we have the following theorem.

Theorem 3. If $k > 0$ and $0 < d < 1$, and $b_{TP} = \text{Bias}(\hat{\alpha}_{TP})$ is the bias of TP, we can say that $\hat{\alpha}_{NTPR}$ is better than the estimator $\hat{\alpha}_{TP}$ using the theorem of MMSE, that is, $MMSE(\hat{\alpha}_{TP}) - MMSE(\hat{\alpha}_{NTPR}) > 0$, if and only if

$$b_{NTPR}^T \left[\sigma^2 \left(A_{TP} \Lambda^{-1} A_{TP}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) + b_{TP} b_{TP}^T \right]^{-1} b_{NTPR} < 1.$$

Proof. Difference between scalar MSE functions,

$$\begin{aligned} MMSE(\hat{\alpha}_{TP}) - MMSE(\hat{\alpha}_{NTPR}) &= \sigma^2 \left(A_{TP} \Lambda^{-1} A_{TP}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) + b_{TP} b_{TP}^T - b_{NTPR} b_{NTPR}^T \\ &= \sigma^2 \text{diag} \left\{ \frac{(\lambda_j + kd)^2}{\lambda_j (\lambda_j + k)^2} - \frac{l_1^2 \lambda_j}{(\lambda_j + k)^2} \right\}_{j=1}^p + b_{TP} b_{TP}^T - b_{NTPR} b_{NTPR}^T, \end{aligned} \quad (65)$$

where $A_{TP} \Lambda^{-1} A_{TP}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T$ is positive definite and we know that it is possible, if and only if $(\lambda_j + kd)^2 - l_1^2 \lambda_j > 0$. For $k > 0$ and $0 < d < 1$, we get that $k^2 d^2 + 2kd\lambda_j + \lambda_j^2 (1 - l_1^2) > 0$. And we can conclude that $A_{TP} \Lambda^{-1} A_{TP}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T$ is a positive definite. By lemma 2, the proof is completed. $\square$

3.1.4. Comparison Between New Proposed Two-Parameter Ridge Estimator and New Biased Estimator Based on Ridge

The difference between $MMSE(\hat{\alpha}_{NTPR})$ and $MMSE(\hat{\alpha}_{NBE})$ is obtained by

$$MMSE(\hat{\alpha}_{NBE}) - MMSE(\hat{\alpha}_{NTPR}) = \sigma^2 A_{NBE} \Lambda^{-1} A_{NBE}^T - \sigma^2 A_{NTPR} \Lambda^{-1} A_{NTPR}^T + b_{NBE} b_{NBE}^T - b_{NTPR} b_{NTPR}^T. \quad (66)$$

If we let $k > 0$, we have the following theorem.

Theorem 4. If $k > 0$ and $0 < d < 1$, and $b_{NBE} = \text{Bias}(\hat{\alpha}_{NBE})$ is the bias of NBE, we can say that $\hat{\alpha}_{NTPR}$ is better than the estimator $\hat{\alpha}_{NBE}$ using the theorem of MMSE, that is, $MMSE(\hat{\alpha}_{NBE}) - MMSE(\hat{\alpha}_{NTPR}) > 0$, if and only if

$$b_{NTPR}^T \left[\sigma^2 \left(A_{NBE} \Lambda^{-1} A_{NBE}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) + b_{NBE} b_{NBE}^T \right]^{-1} b_{NTPR} < 1.$$

Proof. Using the difference between scalar MSE functions,

$$\begin{aligned} MMSE(\hat{\alpha}_{NBE}) - MMSE(\hat{\alpha}_{NTPR}) &= \sigma^2 \left(A_{NBE} \Lambda^{-1} A_{NBE}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) + b_{NBE} b_{NBE}^T - b_{NTPR} b_{NTPR}^T \\ &= \sigma^2 \text{diag} \left\{ \frac{\lambda_j (\lambda_j + (d+k))^2}{(\lambda_j + 1)^2 (\lambda_j + k)^2} - \frac{l_1^2 \lambda_j}{(\lambda_j + k)^2} \right\}_{j=1}^p + b_{NBE} b_{NBE}^T - b_{NTPR} b_{NTPR}^T, \end{aligned} \quad (67)$$

where $A_{NBE} \Lambda^{-1} A_{NBE}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T$ is positive definite and we know that it is possible if and only if $\lambda_j (\lambda_j + (d+k))^2 - l_1^2 \lambda_j (\lambda_j + 1)^2 > 0$. For $k > 0$ and $0 < d < 1$, we get that $(1 - l_1^2) \lambda_j + (k + d - l_1) > 0$, and we can conclude that $A_{NBE} \Lambda^{-1} A_{NBE}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T$ is a positive definite. By lemma 2, the proof is completed. $\square$

3.1.5. Comparison Between Proposed Two-Parameter Ridge Estimator and Yang and Chang Two-Parameter Estimator

The difference between $MMSE(\hat{\alpha}_{NTPR})$ and $MMSE(\hat{\alpha}_{YC})$ is obtained by

$$MMSE(\hat{\alpha}_{YC}) - MMSE(\hat{\alpha}_{NTPR}) = \sigma^2 Y_{k,d} \Lambda^{-1} Y_{k,d}^T - \sigma^2 A_{NTPR} \Lambda^{-1} A_{NTPR}^T + b_{YC} b_{YC}^T - b_{NTPR} b_{NTPR}^T. \quad (68)$$

If we let $k > 0$, we have the following theorem.

Theorem 5. If $k > 0$ and $0 < d < 1$, and $b_{YC} = \text{Bias}(\hat{\alpha}_{YC})$ is the bias of YC , we can say that $\hat{\alpha}_{NTPR}$ is better than the estimator $\hat{\alpha}_{YC}$ using the theorem of MMSE, that is, $MMSE(\hat{\alpha}_{YC}) - MMSE(\hat{\alpha}_{NTPR}) > 0$, if and only if

$$b_{NTPR}^T \left[\sigma^2 \left(Y_{k,d} \Lambda^{-1} Y_{k,d}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) + b_{YC} b_{YC}^T \right]^{-1} b_{NTPR} < 1.$$

Proof. Using the difference between scalar MSE functions,

$$\begin{aligned} MMSE(\hat{\alpha}_{YC}) - MMSE(\hat{\alpha}_{NTPR}) &= \sigma^2 \left(Y_{k,d} \Lambda^{-1} Y_{k,d}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) + b_{YC} b_{YC}^T - b_{NTPR} b_{NTPR}^T \\ &= \sigma^2 \text{diag} \left\{ \frac{\lambda_j (\lambda_j + d)^2}{(\lambda_j + 1)^2 (\lambda_j + k)^2} - \frac{l_1^2 \lambda_j}{(\lambda_j + k)^2} \right\}_{j=1}^p + b_{YC} b_{YC}^T - b_{NTPR} b_{NTPR}^T, \end{aligned} \quad (69)$$

where $Y_{k,d} \Lambda^{-1} Y_{k,d}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T$ is positive definite and we know that it is possible if and only if $\lambda_j (\lambda_j + d)^2 - l_1^2 \lambda_j (\lambda_j + 1)^2 > 0$. For $k > 0$ and $0 < d < 1$, we get that $(1 - l_1) \lambda_j + (d - l_1) > 0$, and we can conclude that $Y_{k,d} \Lambda^{-1} Y_{k,d}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T$ is a positive definite. By lemma 2, the proof is completed. $\square$

4. Discussion

A Monte Carlo simulation is conducted in this section to see the performance of these estimators, following the framework of Hoque and Kibria (2024) [26]. In Section 4.1, we will provide an overview of the simulation technique. The simulation results will be discussed in Section 4.2. We will analyze the simulated data and analyze the estimators' performance across multiple scenarios, and in Section 4.3, real-life data has been collected.

4.1. Simulation Technique

In this section, we did a Monte Carlo simulation experiment, which follows the frameworks of McDonald and Galarneau (1975) [27] and Gibbons (1981) [28]. The explanatory variables are generated according to the following model:

$$X_{ij} = (1 - \rho^2)^{1/2} z_{ij} + \rho z_{i(p+1)}, i = 1, 2, \dots, n, j = 1, 2, \dots, p, \quad (70)$$

where z_{ij} are independent standard normal pseudo-random variables and it has mean zero and the unit variance. The parameter ρ is the correlation among the explanatory variables, while p denotes the total number of predictors. All variables have been standardized so that $X^T X$ and $X^T y$ are expressed in correlation form.

The response variable is generated according to the following:

$$Y_i = \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{ip} + \epsilon_i, \quad (71)$$

where the error ϵ_i is independently normally distributed $(0, \sigma^2)$.

The simulation considers multiple numbers of sample sizes, $n = 30, 50, 100$, and 200 , and different explanatory variables $p = 3, 5$, and 10 . Also, various numbers of correlation levels $\rho = 0.80, 0.90, 0.95$, and 0.99 have been chosen, with three values for error standard deviations: $\sigma = 1, 5$, and 10 . The simulation is replicated 5000 times, using R software version 4.4.0.

Now, the parameter values $\beta_1, \beta_2, \dots, \beta_p$ are chosen from the normalized eigen vectors based on the corresponding largest eigen value of $X^T X$, ensuring that $\beta^T \beta = 1$, and the

performance of the estimators has been evaluated using the estimated MSE, which is defined as follows:

$$MSE(\hat{\beta}) = \frac{1}{5000} \sum_{j=1}^{5000} (\hat{\beta}_{ij} - \beta_i)^T (\hat{\beta}_{ij} - \beta_i) \quad (72)$$

where $\hat{\beta}_{ij}$ denotes the estimate of the i th parameter in the j th replication and β_i is the true parameter value.

Because of the different values of n , p , σ and ρ , the estimated MSE of the estimators are summarized in the corresponding tables. The simulated MSE values are presented in Table 1(1)–(12). In Table 1(1) is $n = 30$ and $p = 3$, in Table 1(2) is $n = 30$ and $p = 5$, in Table 1(3) is $n = 30$ and $p = 10$, in Table 1(4) is $n = 50$ and $p = 3$, in Table 1(5) is $n = 50$ and $p = 5$, in Table 1(6) is $n = 50$, and $p = 10$, in Table 1(7) is $n = 100$, and $p = 3$, in Table 1(8) is $n = 100$ and $p = 5$, in Table 1(9) is $n = 100$ and $p = 10$, in Table 1(10) is $n = 200$ and $p = 3$, in Table 1(11) is $n = 200$ and $p = 5$, and in Table 1(12) is $n = 200$, and $p = 10$. Figure 1a,b depict average MSE trends across ρ and σ .

Table 1. (1) Estimated mean squared error (MSE) values for $n = 30$ and $p = 3$. (2) Estimated mean squared error (MSE) values for $n = 30$ and $p = 5$. (3) Estimated mean squared error (MSE) values for $n = 30$ and $p = 10$. (4) Estimated mean squared error (MSE) values for $n = 50$ and $p = 3$. (5) Estimated mean squared error (MSE) results for $n = 50$ and $p = 5$. (6) Estimated mean squared error (MSE) results for $n = 50$ and $p = 10$. (7) Estimated mean squared error (MSE) results for $n = 100$ and $p = 3$. (8) Estimated mean squared error (MSE) results for $n = 100$ and $p = 5$. (9) Estimated mean squared error (MSE) results for $n = 100$ and $p = 10$. (10) Estimated mean squared error (MSE) results for $n = 200$ and $p = 3$. (11) Estimated mean squared error (MSE) results for $n = 200$ and $p = 5$. (12) Estimated mean squared error (MSE) results for $n = 200$ and $p = 10$.

(1)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	0.3063	1.5081	3.0332	0.6291	3.1221	6.0201	1.2716	6.2526	12.5995	6.7776	34.0651	67.4631	11.9207
LTE	0.0609	0.2799	0.546	0.1046	0.5258	0.982	0.2041	0.9967	1.9156	0.2307	1.0653	2.0367	0.7457
TP1	0.1762	0.8981	1.8142	0.2927	1.5876	3.0293	0.4939	2.5836	5.0977	1.4405	6.3091	11.4264	2.9291
TP2	0.1355	0.5142	0.9762	0.1839	0.7841	1.4025	0.2573	1.1328	2.0941	0.5677	2.2215	3.7186	1.1657
NBE1	0.0687	0.3147	0.6057	0.0999	0.4821	0.8717	0.1497	0.6991	1.2929	0.3294	1.3859	2.3767	0.723
NBE2	0.0877	0.3228	0.595	0.1155	0.4695	0.8238	0.1595	0.663	1.2039	0.3273	1.3629	2.3288	0.705
YC1	0.3776	0.5355	0.6348	0.425	0.5487	0.6475	0.451	0.5362	0.6403	0.4252	0.4089	0.4929	0.5103
YC2	0.1738	0.3848	0.6023	0.2154	0.4703	0.7205	0.2699	0.5351	0.8679	0.3585	0.9055	1.4513	0.5796
MAULE	0.8139	0.8668	0.8859	0.8372	1.0519	0.87	1.1925	2.1992	1.5718	0.6509	0.618	3.7531	1.2759
MAUTP	0.0783	0.2106	0.3665	0.0843	0.2565	0.4258	0.1009	0.3108	0.5477	0.1495	0.592	1.0589	0.3485
MRT	0.0882	0.3132	0.5727	0.1156	0.4561	0.7901	0.1558	0.6116	1.0965	0.261	1.0514	1.7923	0.6087
DK	0.0935	0.4444	0.8519	0.1506	0.7099	1.2824	0.2427	1.1238	2.1277	0.7931	3.6511	6.874	1.5288
GTP	0.2488	0.4905	0.7461	0.2738	0.6389	1.0014	0.3055	0.8394	1.4199	0.4832	1.5694	2.516	0.8777
MLRT	0.0854	0.3766	0.7072	0.1195	0.3972	0.7068	0.215	0.3311	0.4876	0.2228	0.2134	0.2468	0.3425
NBR	1.0006	1.1145	1.114	0.9497	1.1163	1.1511	0.9028	1.1342	1.2153	1.0252	1.5997	1.9209	1.187
BTP	0.0976	0.222	0.3515	0.1275	0.316	0.5083	0.1524	0.4368	0.7265	0.1965	0.4314	0.7799	0.3622
NTP	0.1397	0.7011	1.3975	0.2388	1.2232	2.3004	0.4121	2.0213	3.9029	1.4059	6.508	12.3467	2.7165
NRT	0.1049	0.4812	0.9279	0.1585	0.7393	1.3373	0.24	1.0574	1.9587	0.5282	2.2273	3.8573	1.1348
LKL	0.0767	0.3588	0.6825	0.0981	0.4637	0.8424	0.1198	0.5475	1.0331	0.1542	0.6038	1.181	0.5135
TPR	0.0129	0.0638	0.1485	0.0118	0.0642	0.158	0.0118	0.0719	0.2092	0.014	0.1898	0.9486	0.1587
NTPR	0.0637	0.1172	0.0698	0.0092	0.0569	0.0621	0.0684	0.1158	0.0725	0.0073	0.0289	0.0152	0.0573

Table 1. Cont.

(2)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	0.6257	3.165	6.4301	1.2983	6.5816	13.0217	2.7155	13.305	26.6884	14.517	74.0654	148.66	25.9228
LTE	0.1114	0.6455	1.3165	0.2615	1.4484	2.8016	0.3661	1.8769	3.8933	0.3219	1.5936	2.9872	1.4687
TP1	0.3999	2.2622	4.7112	0.7076	4.1948	8.43	1.3281	7.1789	14.8461	4.5932	23.9541	46.6776	9.9403
TP2	0.1491	0.7454	1.5345	0.2208	1.2479	2.3199	0.3832	1.7264	3.5056	0.8562	4.3362	7.909	2.0779
NBE1	0.0764	0.4907	1.0172	0.1273	0.8306	1.566	0.2263	1.0977	2.3037	0.4299	2.2942	4.1453	1.2171
NBE2	0.1084	0.4117	0.8083	0.1401	0.6645	1.1944	0.2113	0.8578	1.7712	0.3867	2.0735	3.7264	1.0295
YC1	0.6501	0.6165	0.6525	0.6301	0.5867	0.6332	0.5881	0.5451	0.6147	0.4884	0.4617	0.5119	0.5816
YC2	0.3353	0.4411	0.6236	0.3655	0.4936	0.7267	0.3728	0.5106	0.8456	0.4163	1.2997	2.0011	0.7027
MAULE	0.9323	0.9185	0.9111	0.9844	0.8505	0.9703	1.5102	1.0701	2.2772	1.055	8.2229	1.8522	1.7962
MAUTP	0.1043	0.2141	0.3791	0.1005	0.2779	0.4768	0.108	0.3276	0.6755	0.1626	0.8752	1.5233	0.4354
MRT	0.1046	0.3842	0.7704	0.1283	0.5807	1.1011	0.1813	0.762	1.6016	0.3392	1.7684	3.2296	0.9126
DK	0.1426	0.8636	1.7951	0.2874	1.6649	3.214	0.6	2.9342	5.9789	2.725	13.6397	26.7576	5.0503
GTP	0.2993	0.5851	0.9806	0.3099	0.8656	1.4301	0.3907	1.1045	2.1339	0.6059	2.6009	4.6468	1.3294
MLRT	0.1476	0.7423	1.4912	0.2568	0.7834	1.4211	0.4666	0.4624	0.5155	0.33	0.3022	0.3207	0.6033
NBR	1.0691	1.0941	1.0829	1.0091	1.0776	1.1032	0.9308	1.0712	1.1275	0.9236	1.2614	1.3514	1.0918
BTP	0.113	0.3364	0.6101	0.1755	0.6162	1.0922	0.2506	0.8459	1.6076	0.2858	0.9651	1.7349	0.7194
NTP	0.3045	1.6822	3.4585	0.5745	3.1333	6.183	1.1214	5.516	11.1722	4.7542	24.0739	47.0179	9.0826
NRT	0.1289	0.7887	1.6682	0.2074	1.3305	2.5901	0.3797	1.9399	4.0686	1.0175	5.4604	10.1333	2.4761
LKL	0.1119	0.6618	1.376	0.1787	1.025	2.0257	0.268	1.3701	2.8298	0.3876	1.8503	3.6236	1.309
TPR	0.0097	0.0507	0.1097	0.0082	0.0442	0.105	0.0079	0.047	0.142	0.0095	0.1211	0.6818	0.1114
NTPR	0.0409	0.0125	0.1066	0.0036	0.0377	0.2387	0.0014	0.0431	0.0073	0.0001	0.0103	0.008	0.0425
(3)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	1.7886	8.8383	17.2506	3.6271	18.2766	36.5075	7.663	38.861	76.8664	42.4446	210.7642	421.0047	73.6577
LTE	0.4064	2.1051	4.191	0.4648	2.4569	4.898	0.4744	2.4665	4.9297	0.4026	1.8126	3.3776	2.3321
TP1	1.235	6.8816	13.794	2.252	12.9157	26.2612	4.39	24.114	47.658	17.3587	87.9132	174.2615	34.9196
TP2	0.1734	0.9522	1.8794	0.2478	1.4814	2.9616	0.443	2.3233	4.316	1.1162	5.2778	10.0484	2.6017
NBE1	0.1302	1.0082	2.1211	0.2063	1.5205	3.1432	0.3236	2.0139	3.9001	0.5276	2.5823	5.0476	1.8771
NBE2	0.138	0.4582	0.879	0.1463	0.6478	1.2985	0.1892	0.9328	1.7772	0.371	1.8421	3.6993	1.0316
YC1	0.8231	0.7139	0.7004	0.7691	0.6516	0.6469	0.6916	0.6023	0.6054	0.5671	0.5498	0.5504	0.656
YC2	0.5218	0.4394	0.5187	0.4811	0.4324	0.5803	0.4292	0.4712	0.6791	0.474	1.2828	2.2797	0.7158
MAULE	0.9557	0.9624	1.0906	0.9094	0.8677	0.8708	0.8384	0.9413	0.8481	0.8048	0.8754	2.0545	1.0016
MAUTP	0.1536	0.1991	0.3265	0.1177	0.2397	0.4416	0.111	0.3395	0.6228	0.1734	0.7855	1.4868	0.4164
MRT	0.1337	0.4245	0.8447	0.1402	0.6433	1.2921	0.1927	0.9837	1.8717	0.4185	2.1014	4.091	1.0948
DK	0.4671	2.5467	5.0703	1.0176	5.2832	10.5433	2.1728	10.9714	21.5731	11.301	56.6025	113.1655	20.0595
GTP	0.3473	0.6814	1.1259	0.3493	0.9411	1.7267	0.4163	1.3793	2.4297	0.6718	2.7961	5.349	1.5178
MLRT	0.4967	1.8326	3.4332	0.7746	0.8029	0.8715	0.7066	0.6431	0.6453	0.5122	0.49	0.48	0.9741
NBR	1.0473	1.0689	1.0483	0.9836	1.0742	1.0584	0.9164	1.068	1.0615	0.9095	1.0849	1.09	1.0343
BTP	0.2876	1.1157	2.0837	0.4318	1.6748	3.1557	0.5536	2.1119	3.851	0.6051	2.2957	4.2787	1.8704
NTP	0.9938	5.1679	10.223	1.8947	9.785	19.5261	3.7542	19.0377	37.3789	17.5938	88.3292	175.4572	32.4285
NRT	0.184	1.442	3.0364	0.3336	2.4586	5.0656	0.6616	4.118	8.0275	2.0989	10.653	20.9337	4.9177
LKL	0.3501	1.8735	3.7655	0.6099	3.1584	6.3151	0.9246	4.7349	9.3806	1.2001	6.0194	11.9797	4.1927
TPR	0.0063	0.0327	0.0686	0.0047	0.0264	0.0624	0.0043	0.0283	0.0836	0.0055	0.0707	0.3778	0.0643

Table 1. Cont.

NTPR	0.0027	0.0123	0.0456	0.0002	0.0829	0.0153	0.0008	0.0583	0.0275	0.0057	0.0131	0.0811	0.0288
(4)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	0.1719	0.8597	1.702	0.3481	1.7179	3.4335	0.7259	3.4724	6.9551	3.8605	19.1456	38.9415	6.7778
LTE	0.041	0.1755	0.3328	0.066	0.3488	0.6684	0.1306	0.6826	1.3168	0.144	0.7052	1.3497	0.4968
TP1	0.1163	0.5874	1.1558	0.1936	1.0331	2.0586	0.3317	1.7247	3.5318	1.1573	5.2085	9.9177	2.2514
TP2	0.0909	0.3674	0.6808	0.1234	0.5837	1.0774	0.1821	0.8409	1.6378	0.5251	2.0243	3.4619	0.9663
NBE1	0.05	0.2233	0.4154	0.0663	0.3723	0.6761	0.1057	0.5405	1.024	0.3181	1.2679	2.1918	0.6043
NBE2	0.058	0.228	0.4132	0.0729	0.3633	0.6446	0.1092	0.5127	0.9542	0.3108	1.224	2.1129	0.5837
YC1	0.2271	0.4173	0.521	0.2938	0.4566	0.5768	0.3571	0.4768	0.5798	0.389	0.4028	0.4976	0.433
YC2	0.0865	0.2496	0.4164	0.1172	0.3254	0.5678	0.1637	0.4152	0.7272	0.3248	0.7411	1.3511	0.4572
MAULE	0.7962	0.8452	0.8659	0.7602	0.8192	0.8153	0.8235	0.7932	0.7787	1.6878	0.6909	0.8477	0.877
MAUTP	0.0463	0.1578	0.2746	0.053	0.2106	0.365	0.0693	0.2595	0.4497	0.1505	0.5286	0.9387	0.292
MRT	0.0567	0.2278	0.41	0.0758	0.3552	0.6308	0.1123	0.491	0.8857	0.2563	0.9829	1.7152	0.5166
DK	0.0652	0.3234	0.6037	0.101	0.5269	0.9639	0.1716	0.7941	1.5143	0.5916	2.6237	4.8623	1.0951
GTP	0.1611	0.3868	0.5674	0.1884	0.5036	0.8117	0.2356	0.6575	1.1577	0.4492	1.4553	2.4474	0.7518
MLRT	0.0612	0.2956	0.5545	0.0831	0.3548	0.7203	0.1179	0.3531	0.649	0.2094	0.215	0.2499	0.322
NBR	1.0167	1.1195	1.1184	0.9738	1.0921	1.1468	0.9298	1.0654	1.1814	0.8983	1.299	1.4987	1.1117
BTP	0.0602	0.1623	0.2512	0.0878	0.2338	0.3831	0.1291	0.3435	0.5575	0.158	0.4722	0.7795	0.3015
NTP	0.0939	0.4663	0.9036	0.1549	0.8065	1.5812	0.2717	1.3385	2.6597	1.0135	4.5926	8.8089	1.8909
NRT	0.0721	0.3517	0.6627	0.108	0.5627	1.0442	0.1746	0.8139	1.5349	0.4856	1.9446	3.4173	0.931
LKL	0.058	0.2832	0.527	0.0784	0.4029	0.7395	0.1044	0.4854	0.9202	0.1563	0.6469	1.2013	0.467
TPR	0.0075	0.0372	0.0745	0.0067	0.033	0.0735	0.0068	0.0352	0.0795	0.0073	0.0582	0.2238	0.0536
NTPR	0.0047	0.0314	0.029	0.0036	0.0043	0.0204	0.0001	0.0163	0.0285	0.0047	0.0281	0.1627	0.0278
(5)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	0.3492	1.7574	3.4937	0.6939	3.546	7.2227	1.5054	7.3119	14.808	8.1116	39.6786	80.9539	14.1194
LTE	0.0633	0.3951	0.7761	0.1461	0.8997	1.8745	0.2291	1.2509	2.5609	0.235	0.9845	1.9379	0.9461
TP1	0.2566	1.3945	2.7922	0.4306	2.5313	5.3321	0.8391	4.7078	9.7554	3.2775	17.0687	34.3771	6.8969
TP2	0.0971	0.5638	1.0946	0.1298	0.8788	1.9008	0.2534	1.4524	2.9397	0.8048	3.7035	7.045	1.7386
NBE1	0.0532	0.3694	0.7173	0.0776	0.5994	1.3031	0.164	0.9789	2.0327	0.4371	2.0922	4.0096	1.0695
NBE2	0.0646	0.3206	0.5955	0.078	0.4813	1.0272	0.1465	0.7706	1.5638	0.3755	1.8149	3.4823	0.8934
YC1	0.5334	0.5313	0.5754	0.5433	0.5257	0.559	0.529	0.487	0.5489	0.4553	0.4349	0.4827	0.5172
YC2	0.1922	0.3214	0.4751	0.2309	0.3796	0.6076	0.265	0.4318	0.7362	0.3471	0.9056	1.7816	0.5562
MAULE	0.9	0.9127	0.8898	0.8904	0.8936	1.142	0.8488	0.8254	1.0499	0.6977	1.4253	0.7894	0.9388
MAUTP	0.0517	0.1723	0.3006	0.0528	0.2155	0.4342	0.0726	0.2942	0.5984	0.1543	0.7315	1.4192	0.3748
MRT	0.0615	0.2937	0.5512	0.0754	0.4312	0.8975	0.1264	0.6562	1.3631	0.3172	1.542	2.955	0.7725
DK	0.0892	0.5898	1.1516	0.1596	1.0408	2.198	0.3634	1.9781	3.9906	1.7555	8.4985	17.0793	3.2412
GTP	0.2155	0.4822	0.7632	0.2276	0.6359	1.2119	0.2937	0.9689	1.8231	0.5842	2.2489	4.1996	1.1379
MLRT	0.0956	0.5598	1.1102	0.1532	0.7544	1.5311	0.2768	0.7094	1.1505	0.318	0.2957	0.2972	0.6043
NBR	1.0986	1.1054	1.0918	1.0467	1.0798	1.1104	0.9918	1.0344	1.1251	0.8854	1.1012	1.2381	1.0757
BTP	0.0689	0.2035	0.3543	0.1172	0.394	0.7278	0.1976	0.6501	1.2309	0.302	0.9863	1.738	0.5809
NTP	0.1882	1.0542	2.0903	0.3291	1.888	3.9101	0.6795	3.5166	7.0963	2.9876	14.4916	29.3537	5.6321
NRT	0.0899	0.588	1.1561	0.1291	0.9233	1.9908	0.2566	1.5507	3.1914	0.868	4.3952	8.5368	1.973
LKL	0.0775	0.5022	0.9815	0.1206	0.7666	1.6097	0.2116	1.1451	2.3383	0.3957	1.9328	3.8374	1.1599

Table 1. Cont.

TPR	0.0056	0.0289	0.0585	0.0046	0.0244	0.0508	0.0043	0.0237	0.0568	0.0049	0.0384	0.147	0.0373
NTPR	0.0018	0.0067	0.0574	0.0004	0.0034	0.0026	0.0061	0.0072	0.013	0.0194	0.0015	0.0094	0.0107
(6)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	0.8603	4.3265	8.6577	1.8011	8.9652	17.824	3.759	18.8764	36.8339	20.4784	103.2537	205.0225	35.8882
LTE	0.2296	1.2931	2.6889	0.2794	1.5259	3.1452	0.2907	1.5434	3.1126	0.3598	1.3929	2.6109	1.5394
TP1	0.6753	3.7299	7.6583	1.2804	7.2409	14.7737	2.4714	13.9799	27.8378	10.7701	55.8242	112.5467	21.5657
TP2	0.1037	0.7817	1.7944	0.1573	1.3627	2.8379	0.3498	2.1673	4.2439	1.166	5.4155	10.6249	2.5838
NBE1	0.088	0.7772	1.7542	0.1517	1.3696	2.947	0.2949	2.0304	4.1497	0.6362	3.0942	6.0568	1.9458
NBE2	0.079	0.3977	0.8914	0.0921	0.669	1.3935	0.1687	0.972	1.9349	0.4092	2.0441	3.9896	1.0868
YC1	0.7846	0.6552	0.6278	0.7413	0.59	0.5747	0.6778	0.5369	0.5409	0.5179	0.4855	0.5146	0.6039
YC2	0.3904	0.3338	0.4364	0.3729	0.3532	0.5233	0.353	0.4188	0.6624	0.3637	0.8799	1.7087	0.5664
MAULE	0.9686	0.942	0.9428	0.9362	1.3308	1.5362	0.8733	0.8157	1.082	0.743	0.7819	10.37	1.7769
MAUTP	0.0806	0.1635	0.3342	0.0627	0.2362	0.4917	0.0749	0.3569	0.6939	0.1865	0.8645	1.7348	0.44
MRT	0.076	0.3352	0.7623	0.0815	0.5572	1.2191	0.137	0.9098	1.8286	0.4107	2.1053	4.2338	1.0547
DK	0.2367	1.4809	3.1566	0.549	3.119	6.3426	1.2325	6.328	12.5349	6.3119	31.3781	62.9752	11.3038
GTP	0.2673	0.5679	1.0826	0.2746	0.8751	1.6302	0.3594	1.2946	2.3958	0.7487	3.0355	5.7682	1.525
MLRT	0.2804	1.4615	2.9726	0.5367	1.8277	3.3865	0.6942	0.611	0.6009	0.4974	0.4598	0.4438	1.1477
NBR	1.0691	1.0846	1.0721	1.027	1.0691	1.0863	0.9448	1.0284	1.0875	0.8521	1.0074	1.0986	1.0356
BTP	0.1504	0.5602	1.0528	0.2845	1.0767	2.0107	0.4277	1.6059	2.8809	0.6491	2.1894	3.7494	1.3865
NTP	0.52	2.8509	5.8494	1.0251	5.4572	11.0372	2.041	10.4892	20.7383	9.5802	47.9265	95.8208	17.778
NRT	0.1166	1.0667	2.4756	0.2089	1.9062	4.1756	0.4607	3.3033	6.7456	1.8246	9.5689	19.2528	4.2588
LKL	0.203	1.2316	2.6122	0.4064	2.23	4.5668	0.7082	3.6739	7.3503	1.2692	6.389	12.8564	3.6248
TPR	0.0036	0.0185	0.0364	0.0026	0.0138	0.0305	0.0024	0.014	0.0327	0.0028	0.0223	0.0969	0.023
NTPR	0.0044	0.0072	0.0075	0.0009	0.0021	0.0032	0.0022	0.012	0.0024	0.0002	0.0023	0.0288	0.0061
(7)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	0.0813	0.4064	0.8168	0.1633	0.8188	1.6289	0.3393	1.7654	3.4602	1.8716	9.1838	18.1613	3.2248
LTE	0.0232	0.092	0.1787	0.04	0.1802	0.3596	0.0734	0.43	0.8006	0.0784	0.3965	0.7538	0.2839
TP1	0.0622	0.3137	0.6208	0.1089	0.5561	1.1164	0.1894	1.0739	2.0939	0.7244	3.619	7.0017	1.4567
TP2	0.0525	0.2231	0.4245	0.0753	0.3402	0.6548	0.1109	0.597	1.1026	0.3586	1.5984	2.8942	0.7027
NBE1	0.0335	0.1366	0.2629	0.0442	0.2134	0.414	0.0645	0.3875	0.704	0.2268	0.9977	1.8087	0.4412
NBE2	0.0355	0.1389	0.2643	0.0459	0.2094	0.4004	0.0647	0.3707	0.6633	0.2198	0.9407	1.6887	0.4202
YC1	0.0698	0.2492	0.3634	0.1228	0.312	0.4244	0.193	0.3744	0.4873	0.31	0.3743	0.491	0.3143
YC2	0.0386	0.1317	0.2398	0.0505	0.1801	0.3292	0.0702	0.272	0.4944	0.1992	0.5076	0.9978	0.2926
MAULE	0.8289	0.8927	0.9049	0.7656	0.832	0.8408	0.7279	0.7681	0.7894	0.6657	0.6131	0.6992	0.7774
MAUTP	0.0282	0.1042	0.1914	0.033	0.135	0.2503	0.0431	0.2121	0.3574	0.1191	0.3948	0.7143	0.2152
MRT	0.0348	0.1402	0.2634	0.0461	0.2093	0.391	0.0684	0.3678	0.6379	0.1993	0.7978	1.4444	0.3834
DK	0.0419	0.1993	0.3874	0.0618	0.309	0.5944	0.1015	0.5495	0.9877	0.3662	1.6678	3.1137	0.6984
GTP	0.0663	0.2495	0.3987	0.0962	0.3224	0.5264	0.1392	0.4847	0.818	0.3172	1.1395	2.0206	0.5482
MLRT	0.0396	0.1893	0.3659	0.0561	0.2765	0.5337	0.0719	0.3674	0.6957	0.173	0.1978	0.2482	0.2679
NBR	1.0302	1.1344	1.1332	0.9761	1.0931	1.1424	0.9489	1.0274	1.1473	0.8724	1.0093	1.2644	1.0649
BTP	0.027	0.0997	0.1657	0.0427	0.1506	0.2451	0.0762	0.2515	0.3913	0.146	0.4578	0.7669	0.235
NTP	0.0552	0.2606	0.5101	0.0902	0.4455	0.8752	0.1535	0.8364	1.5956	0.6071	2.848	5.4298	1.1423
NRT	0.0442	0.2116	0.412	0.0665	0.3366	0.6497	0.1062	0.5891	1.0687	0.3486	1.4717	2.6987	0.667

Table 1. Cont.

LKL	0.0394	0.1858	0.3612	0.054	0.27	0.5172	0.0778	0.4184	0.7529	0.1435	0.6233	1.1716	0.3846
TPR	0.0035	0.0178	0.0357	0.0033	0.0161	0.0324	0.0032	0.0159	0.0335	0.0033	0.019	0.0497	0.0195
NTPR	0.0002	0.0094	0.0088	0.0015	0.0094	0.0281	0.0001	0.006	0.004	0.0006	0.0516	0.0593	0.0149
(8)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	0.1626	0.802	1.6638	0.3311	1.6414	3.355	0.6947	3.4427	6.9683	3.806	18.5566	37.1572	6.5485
LTE	0.0367	0.1964	0.4165	0.0751	0.4631	1.0092	0.1186	0.7066	1.4589	0.1718	0.5987	1.1302	0.5318
TP1	0.1374	0.6902	1.436	0.2426	1.2943	2.7157	0.446	2.515	5.2568	1.9453	10.3086	20.8816	3.9891
TP2	0.0637	0.3302	0.6986	0.078	0.5115	1.1555	0.1292	0.9484	2.0149	0.6317	2.7951	5.486	1.2369
NBE1	0.036	0.2081	0.4481	0.0495	0.3551	0.8074	0.0885	0.6809	1.4432	0.3917	1.7603	3.4904	0.8133
NBE2	0.038	0.1865	0.3925	0.0433	0.2884	0.6535	0.0711	0.5408	1.1293	0.333	1.4302	2.8273	0.6612
YC1	0.3548	0.4195	0.4627	0.3954	0.4132	0.456	0.4196	0.4107	0.454	0.3731	0.3844	0.4756	0.4183
YC2	0.0785	0.18	0.3144	0.0985	0.2268	0.4107	0.13	0.3113	0.5639	0.2513	0.5654	1.0795	0.3509
MAULE	0.8978	0.9054	0.9097	0.8629	0.8599	0.8677	0.9539	0.9302	0.7897	0.7199	0.7008	1.2243	0.8852
MAUTP	0.0264	0.109	0.2226	0.0248	0.1403	0.308	0.035	0.2309	0.4653	0.1363	0.5543	1.1207	0.2811
MRT	0.0372	0.1752	0.3605	0.0409	0.258	0.5678	0.0667	0.4586	0.9497	0.2631	1.2015	2.3568	0.5613
DK	0.0579	0.324	0.6842	0.0851	0.5532	1.2138	0.1768	1.1137	2.3119	1.0062	4.7859	9.5208	1.8195
GTP	0.1168	0.3212	0.525	0.1452	0.4113	0.7718	0.1829	0.6549	1.2704	0.4711	1.7361	3.3363	0.8286
MLRT	0.0585	0.3268	0.6834	0.093	0.5288	1.1123	0.1595	0.7336	1.5017	0.2733	0.2752	0.3068	0.5044
NBR	1.1419	1.1257	1.1054	1.1074	1.0996	1.1162	1.0515	1.0363	1.1129	0.9005	0.9437	1.126	1.0723
BTP	0.0335	0.1097	0.1939	0.0609	0.2119	0.391	0.1213	0.406	0.7567	0.2767	0.9821	1.6877	0.436
NTP	0.1042	0.5346	1.1149	0.1768	0.9683	2.0433	0.3379	1.8725	3.8677	1.6398	7.9057	15.7273	3.0244
NRT	0.0623	0.3405	0.7139	0.0813	0.5392	1.193	0.1394	0.9806	2.0817	0.6362	3.1108	6.2068	1.3405
LKL	0.0539	0.2996	0.6315	0.0738	0.472	1.0302	0.1331	0.8141	1.6883	0.3595	1.8005	3.6046	0.9134
TPR	0.0028	0.0134	0.0271	0.0023	0.0112	0.023	0.0022	0.0108	0.0222	0.0022	0.0122	0.0321	0.0135
NTPR	0.0009	0.011	0.007	0.0014	0.0005	0.0008	0.0004	0.0058	0.0445	0.0002	0.0018	0.0063	0.0067
(9)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	0.3755	1.9144	3.7812	0.7794	3.8929	7.788	1.6424	8.0984	16.2122	8.8673	44.2022	88.7141	15.5223
LTE	0.1134	0.6636	1.377	0.139	0.8037	1.6762	0.1387	0.8012	1.6527	0.2865	0.9699	1.9161	0.8782
TP1	0.3295	1.748	3.5017	0.6162	3.391	6.9603	1.1896	6.7079	13.6723	5.5947	29.6761	60.335	11.1435
TP2	0.0624	0.4768	1.1025	0.0761	0.8456	1.9818	0.1533	1.5995	3.2339	0.9143	4.3305	9.0925	1.9891
NBE1	0.0604	0.4603	1.0292	0.0926	0.8952	2.038	0.1676	1.6233	3.3802	0.6109	3.1323	6.6594	1.6791
NBE2	0.0391	0.2476	0.5754	0.042	0.4537	1.0707	0.0783	0.8516	1.6867	0.3689	1.8116	3.9461	0.931
YC1	0.6591	0.5869	0.5566	0.6685	0.5196	0.4948	0.6251	0.4791	0.4566	0.4776	0.4036	0.4269	0.5295
YC2	0.1825	0.2036	0.2934	0.215	0.231	0.371	0.2247	0.2981	0.504	0.2574	0.5679	1.07	0.3682
MAULE	0.9518	0.9406	0.9337	0.9521	0.91	1.1561	0.9164	0.8272	0.8015	0.7436	0.6632	0.8696	0.8888
MAUTP	0.0298	0.1038	0.238	0.025	0.1687	0.3922	0.0312	0.2937	0.6021	0.1526	0.7456	1.6529	0.3696
MRT	0.0373	0.2054	0.4767	0.0378	0.3533	0.8338	0.0601	0.6345	1.3505	0.3164	1.7368	3.7161	0.8132
DK	0.1115	0.7351	1.5754	0.243	1.5509	3.2741	0.554	3.2346	6.4453	3.1675	15.3463	30.7836	5.5851
GTP	0.1838	0.3872	0.6923	0.1887	0.572	1.1651	0.2151	0.9882	1.8454	0.6156	2.4261	4.9678	1.1873
MLRT	0.1419	0.8201	1.6757	0.2826	1.4329	2.8446	0.5092	1.8244	3.2902	0.4763	0.4175	0.41	1.1771
NBR	1.1045	1.0926	1.0852	1.0832	1.0656	1.1044	1.0228	0.9942	1.082	0.8456	0.9157	1.0434	1.0366
BTP	0.0645	0.2467	0.4652	0.1464	0.5781	1.0815	0.2677	1.0313	1.9262	0.57	1.9105	3.2562	0.962
NTP	0.2498	1.3624	2.7586	0.4653	2.5897	5.3178	0.9289	5.0593	10.1194	4.6387	22.7344	45.6582	8.4902

Table 1. Cont.

NRT	0.0811	0.6212	1.3926	0.1095	1.112	2.5586	0.2136	2.0981	4.422	1.2169	6.8241	14.4233	2.9228
LKL	0.1039	0.6728	1.4362	0.2114	1.3025	2.7293	0.418	2.3323	4.6998	1.1672	5.8616	11.749	2.7237
TPR	0.0017	0.0087	0.0176	0.0013	0.0066	0.0136	0.0012	0.0059	0.0125	0.0011	0.007	0.0197	0.0081
NTPR	0.0005	0.0126	0.0325	0.0034	0.0013	0.0306	0.0003	0.0004	0.0117	0.0005	0.0006	0.0056	0.0083
(10)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	0.0404	0.2053	0.4065	0.0789	0.3985	0.8219	0.1668	0.8266	1.6606	0.9111	4.5839	9.1765	1.6064
LTE	0.0141	0.0502	0.094	0.022	0.1016	0.1989	0.0464	0.2266	0.4489	0.0427	0.212	0.4208	0.1565
TP1	0.0331	0.1733	0.3375	0.0587	0.3036	0.6185	0.1097	0.5612	1.1382	0.4106	2.2948	4.6083	0.8873
TP2	0.0302	0.142	0.261	0.0448	0.21	0.4057	0.0717	0.34	0.664	0.2174	1.1384	2.2117	0.4781
NBE1	0.0222	0.0928	0.1619	0.0299	0.1296	0.252	0.0442	0.2162	0.4263	0.1394	0.7362	1.4183	0.3058
NBE2	0.0228	0.0942	0.1637	0.0305	0.1288	0.2471	0.044	0.2085	0.4073	0.1338	0.6901	1.3152	0.2905
YC1	0.0231	0.127	0.2211	0.0366	0.1791	0.2846	0.0836	0.2361	0.3536	0.2045	0.3463	0.4696	0.2138
YC2	0.0234	0.0898	0.1447	0.0316	0.109	0.1873	0.0439	0.1433	0.2798	0.1006	0.3775	0.8079	0.1949
MAULE	0.8235	0.9059	0.935	0.7819	0.8709	0.8928	0.7417	0.8166	0.8309	0.678	0.6632	0.6548	0.7996
MAUTP	0.02	0.0813	0.137	0.0243	0.0965	0.1743	0.0324	0.1323	0.2526	0.075	0.3261	0.5953	0.1623
MRT	0.0228	0.0981	0.1729	0.0309	0.1331	0.25	0.0465	0.2099	0.4017	0.1318	0.6302	1.1748	0.2752
DK	0.0263	0.128	0.2407	0.0387	0.1928	0.3784	0.0634	0.3151	0.6099	0.2222	1.0709	2.0648	0.4459
GTP	0.026	0.1553	0.2609	0.039	0.2003	0.3408	0.0702	0.2916	0.5063	0.199	0.8254	1.5546	0.3725
MLRT	0.0257	0.1246	0.2328	0.036	0.1833	0.356	0.0549	0.2806	0.5298	0.1084	0.2908	0.5042	0.2273
NBR	0.9779	1.1412	1.1567	0.9103	1.0886	1.1506	0.8936	1.0322	1.1351	0.8494	0.9311	1.08	1.0289
BTP	0.0167	0.0656	0.1094	0.0201	0.0922	0.1645	0.0384	0.1553	0.2548	0.1235	0.395	0.6767	0.176
NTP	0.032	0.1542	0.2912	0.0529	0.254	0.5049	0.0927	0.4493	0.893	0.3413	1.7496	3.4656	0.6901
NRT	0.0269	0.1317	0.2491	0.0409	0.2056	0.4057	0.0679	0.3434	0.6693	0.2194	1.0681	2.0425	0.4559
LKL	0.0255	0.1235	0.2322	0.0361	0.1792	0.3518	0.0551	0.273	0.5277	0.12	0.5681	1.1027	0.2996
TPR	0.0018	0.0085	0.0177	0.0015	0.0079	0.0159	0.0016	0.0078	0.0153	0.0016	0.0084	0.018	0.0088
NTPR	0.0013	0.0281	0.0045	0.0031	0.0003	0.0024	0.0048	0.0021	0.0037	0.0009	0.0107	0.0042	0.0055
(11)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	0.0785	0.3963	0.8037	0.1569	0.818	1.6457	0.3297	1.6756	3.373	1.8443	9.2159	18.787	3.2604
LTE	0.024	0.1126	0.2293	0.0397	0.2524	0.5346	0.0614	0.3607	0.7804	0.0823	0.3827	0.7366	0.2997
TP1	0.0717	0.3603	0.7283	0.1303	0.6951	1.4099	0.2405	1.3155	2.7198	1.0158	5.946	12.5296	2.2636
TP2	0.0429	0.2089	0.4163	0.053	0.3222	0.6834	0.0743	0.532	1.1825	0.3097	1.9775	4.1669	0.8308
NBE1	0.0249	0.1262	0.2612	0.0325	0.2191	0.4723	0.0518	0.3907	0.8728	0.2137	1.3864	2.8884	0.5783
NBE2	0.0253	0.1188	0.2388	0.028	0.1834	0.3957	0.0389	0.3082	0.6939	0.1732	1.0898	2.2435	0.4615
YC1	0.1962	0.2981	0.3582	0.2481	0.3156	0.3489	0.298	0.3109	0.3567	0.3054	0.3232	0.3753	0.3112
YC2	0.0335	0.1074	0.1897	0.0406	0.1362	0.2595	0.0557	0.1909	0.3616	0.1312	0.4252	0.7967	0.2274
MAULE	0.9227	0.9416	0.9456	0.8765	0.8879	0.8888	0.8344	0.8275	0.8302	0.8725	0.7945	0.6422	0.8554
MAUTP	0.019	0.0814	0.1572	0.0166	0.1003	0.2152	0.0192	0.1427	0.3161	0.0706	0.4231	0.8429	0.2004
MRT	0.0261	0.1174	0.2308	0.0279	0.1664	0.3553	0.0369	0.2647	0.5741	0.1418	0.8769	1.796	0.3845
DK	0.0403	0.1985	0.4052	0.0519	0.3202	0.6772	0.088	0.5796	1.2609	0.4881	2.7214	5.5695	1.0334
GTP	0.0429	0.2101	0.3382	0.0658	0.2742	0.4861	0.104	0.401	0.7766	0.2634	1.2503	2.5617	0.5645
MLRT	0.0389	0.1916	0.3952	0.0537	0.3279	0.682	0.0964	0.5317	1.1169	0.2392	0.3116	0.3856	0.3642
NBR	1.1649	1.1514	1.1227	1.1469	1.1303	1.1345	1.1195	1.0747	1.1149	0.9914	0.8993	0.999	1.0875
BTP	0.0162	0.0667	0.1176	0.0267	0.1218	0.2197	0.0626	0.2422	0.4409	0.2218	0.8071	1.4859	0.3191

Table 1. Cont.

NTP	0.0587	0.2922	0.5902	0.0971	0.5315	1.0858	0.1741	0.9753	2.0252	0.8182	4.3675	8.9104	1.6605
NRT	0.0422	0.2088	0.4241	0.0554	0.3366	0.7026	0.0814	0.5575	1.2183	0.3209	2.0821	4.3742	0.867
LKL	0.0388	0.1908	0.3897	0.0482	0.2954	0.6235	0.0762	0.4908	1.0633	0.2613	1.4396	2.9203	0.6532
TPR	0.0013	0.0068	0.0136	0.0011	0.0056	0.0113	0.001	0.0051	0.0105	0.001	0.0052	0.0116	0.0062
NTPR	0.0029	0.0122	0.0048	0.0012	0.0007	0.0019	0.0003	0.0004	0.0401	0.0005	0.0149	0.0003	0.0067
(12)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	0.1773	0.8933	1.7782	0.3707	1.8411	3.6662	0.7845	3.876	7.6647	4.2855	21.2913	42.7157	7.4454
LTE	0.0578	0.3385	0.712	0.0715	0.4118	0.8727	0.0694	0.4309	0.8799	0.1719	0.8084	1.4128	0.5198
TP1	0.1648	0.843	1.6882	0.3182	1.671	3.3787	0.6103	3.3801	6.8204	2.8412	16.121	32.9772	5.9012
TP2	0.0454	0.2944	0.6524	0.0488	0.4794	1.1107	0.0681	0.9261	2.0588	0.4902	3.4087	6.6854	1.3557
NBE1	0.0413	0.2671	0.5836	0.0654	0.5176	1.1627	0.1009	1.0256	2.244	0.4173	3.0533	6.13	1.3007
NBE2	0.0243	0.1528	0.349	0.0243	0.2609	0.6197	0.0322	0.5319	1.183	0.2449	1.686	3.2509	0.6967
YC1	0.5138	0.4826	0.4876	0.5324	0.4386	0.4307	0.5392	0.4099	0.3881	0.4441	0.3385	0.3501	0.4463
YC2	0.0672	0.1135	0.1807	0.0849	0.1292	0.2254	0.1041	0.1766	0.3408	0.1468	0.4517	0.8338	0.2379
MAULE	0.9525	0.9582	0.962	0.9583	0.911	1.04	0.9304	0.8725	0.8641	1.1409	0.6678	0.6478	0.9088
MAUTP	0.0143	0.0713	0.1585	0.011	0.0994	0.2424	0.0119	0.1856	0.4296	0.0866	0.6504	1.2697	0.2692
MRT	0.0237	0.1335	0.2922	0.0221	0.2029	0.4709	0.0278	0.3721	0.856	0.1719	1.3555	2.7412	0.5558
DK	0.0605	0.3815	0.8192	0.1181	0.7662	1.6418	0.2575	1.6567	3.4056	1.5696	8.2544	16.3268	2.9382
GTP	0.0977	0.2591	0.4477	0.1214	0.3588	0.6801	0.1317	0.6065	1.217	0.3699	1.9679	3.7767	0.8362
MLRT	0.0718	0.4265	0.8974	0.1464	0.8261	1.7008	0.29	1.4342	2.8149	0.4642	0.3748	0.3643	0.8176
NBR	1.141	1.1114	1.0922	1.1345	1.097	1.1075	1.1048	1.0248	1.0846	0.9391	0.8543	0.956	1.0539
BTP	0.0294	0.1154	0.222	0.0734	0.2997	0.5791	0.1589	0.6389	1.2234	0.4424	1.5503	2.7994	0.6777
NTP	0.1282	0.6772	1.3694	0.2341	1.2821	2.6253	0.4496	2.5546	5.1483	2.2883	11.9138	23.7428	4.3678
NRT	0.0597	0.3675	0.791	0.0736	0.6166	1.3933	0.1074	1.184	2.6184	0.6288	4.693	9.6195	1.8461
LKL	0.0584	0.366	0.7838	0.1105	0.7001	1.4959	0.226	1.3811	2.8371	0.8641	4.5329	9.0525	1.8674
TPR	0.0009	0.0043	0.0086	0.0006	0.0031	0.0064	0.0006	0.0029	0.0058	0.0005	0.0029	0.0066	0.0036
NTPR	0.0009	0.0058	0.0115	0.0001	0.0008	0.0034	0.0035	0.0038	0.0086	0.0004	0.0115	0.0543	0.0087

All simulations and numerical computations were performed using the R version 4.4.0.

4.2. Discussion of Simulation Results

Table 1 summarizes the results of the simulated mean squared error (MSE) outcomes of these estimators under different controlled settings, considering various sample sizes $n = 30, 50, 100, 200$ and predictors $p = 3, 5, 10$. These tables report MSE values across different correlation levels ($\rho = 0.80, 0.90, 0.95, 0.99$) and varying error standard deviations ($\sigma = 1, 5, 10$). The OLS estimator consistently exhibits the largest MSE, confirming its vulnerability under multicollinearity. Across all scenarios, a general trend is observed where increasing the sample size results in lower MSE values. In contrast, higher correlation levels and higher error standard deviations tend to produce larger MSE values.

An examination of Table 1 also reveals that increasing the number of explanatory variables generally leads to higher MSE values. For instance, Table 1(2) presents the MSE results for $n = 30$ and $p = 5$, where an increase in the correlation levels corresponds to a rise in MSE. Among the compared estimators, YC1 and TPR consistently yield lower MSE values throughout the simulation. The newly proposed NTPR estimator consistently yielded the lowest or near-lowest MSE values across almost all conditions, particularly when $\rho \geq 0.95$ and $\sigma \geq 5$.

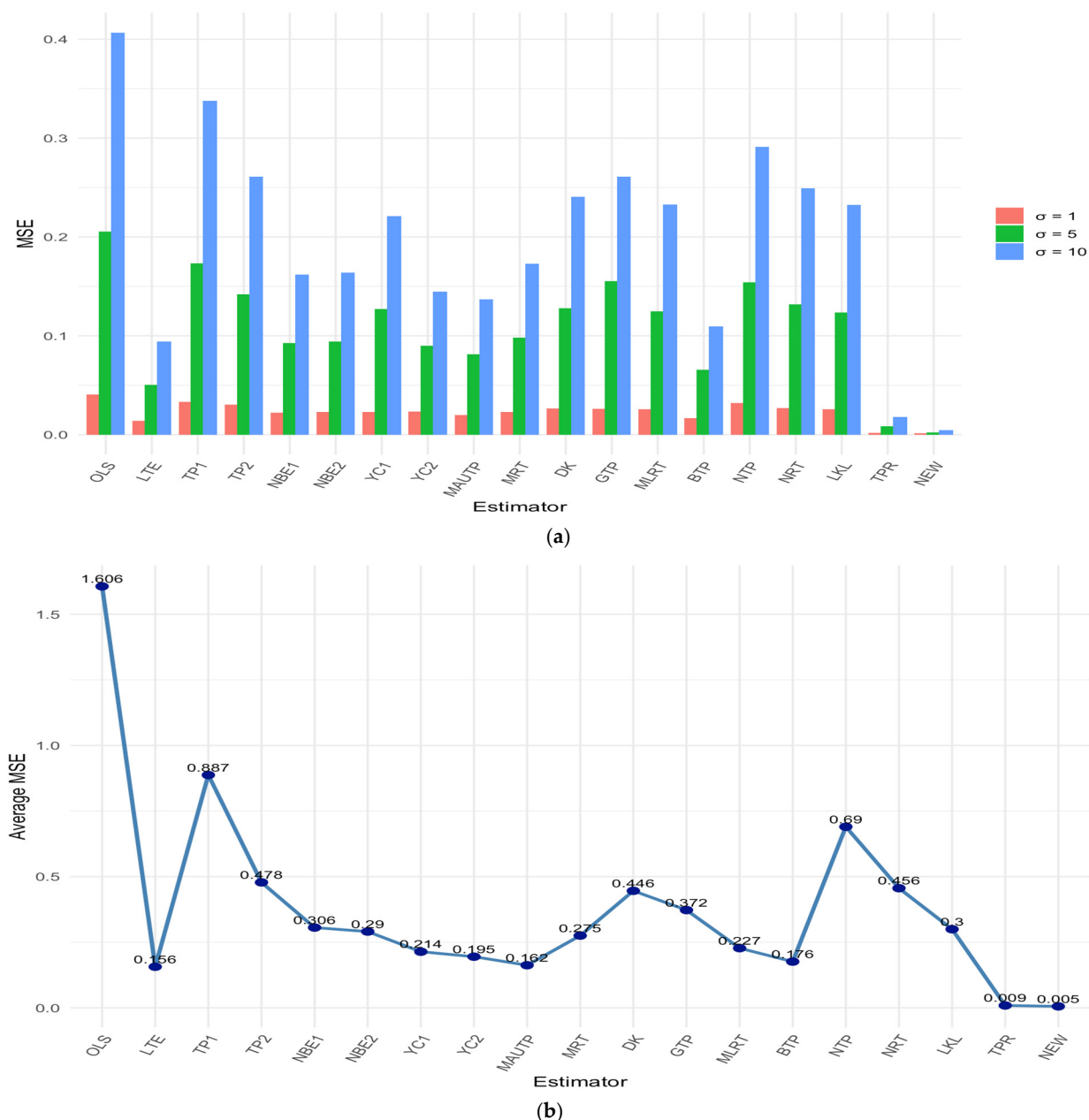


Figure 1. (a) Bar plot for MSE value comparison by σ for fixed $n = 200$, $p = 3$ and $\rho = 0.80$. (b). Trend comparison for average MSE by ρ and σ for fixed $n = 30$ and $p = 3$.

For better visualization, we also plotted the relationship between MSE by correlation $\rho = 0.80$ for $n = 200$ and $p = 3$ in Figure 1a, as well as the average trends between MSE, correlation ρ and error standard deviation σ for fixed $n = 30$ and $p = 3$ in Figure 1b. Both figures confirm the analyses presented in Table 1(1)–(12).

Figure 1 visually summarizes the simulation results reported in Table 1. Figure 1a displays the bar plot of average MSE values across correlation levels $\rho = 0.80, 0.90, 0.95, 0.99$ for fixed $\sigma = 1, 5, 10$. As ρ increases, the MSE values of OLS grow exponentially, confirming its instability under strong collinearity. By contrast, the NTPR estimator shows only a mild, nearly flat increase in MSE, remaining the lowest across all noise levels. This pattern underscores the estimator's ability to balance bias and variance effectively, even as the explanatory variables become highly correlated.

Figure 1b illustrates the trend of average MSE as a function of ρ and σ for $n = 100$ and $p = 10$. Here, each line represents one estimator's mean MSE trajectory. The slope

of the NTPR curve is markedly smaller than that of competing estimators, demonstrating its strong resistance to multicollinearity and noise amplification. The consistency of this behavior across different parameter configurations provides empirical validation of the theoretical dominance conditions established in Section 3.1.

4.2.1. Estimator Performance as a Function of n

An analysis of Table 1 compares the estimated MSE values for various estimators across different sample sizes ($n = 30, 50, 100$, and 200), while accounting for different numbers of regressors (p), correlation levels (ρ) and error standard deviation (σ). In general, as the number of sample sizes (n) increases, the estimated MSE tends to decrease across different combinations of p , ρ and σ . Among the estimators, NTPR yields the lowest MSE values, followed closely by YC1 and TPR, which also demonstrate strong performance, and the OLS estimator shows the highest MSE values.

4.2.2. Estimator Performance as a Function of p

Table 1 presents a comparative analysis of the estimated MSE values for various estimators across the values of p (3, 5, and 10), considering multiple sample sizes (n), correlation levels (ρ) and error standard deviation σ . Generally, an increase in the number of predictors (p) is associated with an increase in the estimated MSE across most settings of n , ρ and σ . Among the estimators evaluated, NTPR achieves the lowest MSE values, with TPR and YC1 also performing well, and the OLS estimator records the highest MSE values.

4.2.3. Estimator Performance as a Function of ρ

Based on the results presented in Table 1, we assess the estimated MSE values of various estimators across the numbers of correlation levels ($\rho = 0.80, 0.90, 0.95$, and 0.99), sample sizes (n), number of regressors (p) and error standard deviation (σ). Generally, an increase in the number of correlation (ρ) is associated with an increase in estimated MSE across different numbers of sample sizes and correlation levels. Among the evaluated estimators, NTPR produces the lowest MSE values, with TPR also demonstrating favorable performance, and OLS yields the highest MSE values overall.

4.2.4. Estimator Performance as a Function of σ

Table 1 provides a comparative assessment of the estimated MSE values for various estimators across different settings of error standard deviation ($\sigma = 1, 5$, and 10), sample sizes (n), number of regressors (p) and correlation levels (ρ). In most cases, increasing the number of deviation (σ) leads to an increase in MSE values across the examined scenarios. Among the considered estimators, NTPR achieves the lowest MSE values, while TPR also performs well, and OLS shows the highest MSE values.

Evaluating the performance of the estimators with respect to sample size (n), number of predictors (p), correlation coefficient (ρ), and error standard deviation (σ) provides valuable insight into their behavior and efficiency. Among the examined methods, NTPR and TPR demonstrate strong performance across varying conditions, whereas the OLS often exhibits relatively higher MSE values. Such comparative analysis is crucial for guiding the selection of suitable estimation techniques in regression modeling.

4.2.5. Estimator Performance Under Outliers

Following Yasmin and Kibria (2025) [29], Table 2(1)–(3) provides a different comparison for fixed $n = 100$; different explanatory variables of $p = 3, 5$, and 10 ; with different correlations of $\rho = 0.8, 0.9, 0.95$, and 0.99 ; and error standard deviation of $1, 5$, and 10 . Also, we generated 25% outliers in this model, using random replacement of some values

from the error vector of the third quartile of the errors plus five times the interquartile range (IQR) of the errors.

Table 2. (1) Estimated MSE values for $n = 100$ and $p = 3$, with 25% outliers. (2) Estimated MSE values for $n = 100$ and $p = 5$, with 25% outliers. (3) Estimated MSE values for $n = 100$ and $p = 10$, with 25% outliers.

(1)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	1.1337	5.8806	11.7277	2.3319	11.9986	23.9126	4.964	25.0182	47.8905	25.958	131.1298	271.4453	46.9492
LTE	0.2443	1.2041	2.357	0.5018	2.532	4.9396	1.1997	5.4032	10.3841	1.0717	5.173	10.4303	3.7867
TP1	0.8495	4.2059	8.3216	1.5739	8.0124	15.7881	3.0512	15.0266	28.584	9.778	49.0534	101.1022	20.4456
TP2	0.5574	2.5294	4.9351	0.8926	4.4464	8.6242	1.631	7.4841	14.2535	3.776	18.3177	38.144	8.7993
NBE1	0.3466	1.5407	2.9578	0.5556	2.7122	5.1124	1.0557	4.5944	8.7964	2.365	11.4845	24.3979	5.4933
NBE2	0.3453	1.4848	2.814	0.5326	2.5376	4.7556	0.9912	4.2087	8.039	2.1985	10.6263	22.6838	5.1015
YC1	0.4241	0.9649	1.4451	0.4875	1.2221	1.851	0.592	1.5	2.4904	0.6018	1.8884	3.6182	1.4238
YC2	0.3159	1.3481	2.5959	0.4382	2.2256	4.3815	0.7472	3.5214	6.6564	1.3631	6.8702	14.9791	3.7869
MAULE	0.906	0.9231	0.9323	0.8516	0.8663	0.987	0.7994	0.8228	0.973	0.7142	0.6622	0.7666	0.8504
MAUTP	0.244	1.0329	1.9793	0.3168	1.5598	2.9122	0.5363	2.2504	4.3853	0.9909	5.4058	12.4854	2.8416
MRT	0.3433	1.5159	2.9308	0.5151	2.4998	4.738	0.936	4.0031	7.5678	1.9271	9.1334	19.3498	4.6217
DK	0.5119	2.2952	4.4438	0.8002	3.9044	7.4045	1.45	6.3949	12.2345	4.1962	20.6177	43.191	8.9537
GTP	0.4747	1.6701	3.1358	0.673	2.8977	5.4195	1.1688	4.889	9.1693	2.6142	12.3997	25.8878	5.8666
MLRT	0.4852	2.2086	4.2947	0.7334	3.7231	7.0652	0.9922	5.0076	9.5033	0.3059	0.9445	1.7845	3.0874
NBR	1.1144	1.0379	1.0247	1.1348	1.0566	1.0406	1.1635	1.0894	1.0839	1.3356	1.6082	2.0837	1.2311
BTP	0.2105	0.7902	1.4601	0.3138	1.345	2.3953	0.5475	2.1416	4.1181	1.04	4.4794	8.9007	2.3119
NTP	0.6906	3.3683	6.6598	1.2176	6.2176	12.1344	2.3376	11.2625	21.4924	7.5289	37.3053	77.4832	15.6415
NRT	0.5467	2.4702	4.7902	0.8794	4.259	8.1099	1.5591	6.961	13.268	3.5663	17.2691	36.2605	8.3283
LKL	0.4773	2.1384	4.1419	0.6944	3.3891	6.3962	1.1033	4.8576	9.3083	1.6204	7.8088	16.3691	4.8587
TPR	0.0498	0.3622	1.4345	0.0453	0.4967	2.3217	0.0499	0.9248	4.8369	0.0963	6.6577	43.6313	5.0756
NTPR	0.0382	0.298	0.3605	0.1559	0.0245	0.0971	0.0101	0.3976	0.1905	0.0353	0.1034	1.1296	0.2367
(2)													
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	2.305	11.7423	23.2288	4.7857	23.821	46.8559	9.8077	50.1667	97.745	53.3967	267.2609	540.5056	94.3018
LTE	0.5865	3.0406	5.8503	1.4713	7.0817	13.9243	2.0787	10.7833	21.4843	1.5927	7.3296	14.4831	7.4755
TP1	1.9895	10.1583	20.0527	3.9122	19.6656	38.679	7.4192	38.8109	76.0002	30.0546	149.8613	302.2821	58.2405
TP2	0.9628	4.9371	9.4678	1.6954	8.3132	16.2082	2.7632	14.5258	27.8733	7.7338	36.294	73.7439	17.0432
NBE1	0.6262	3.1883	6.0606	1.1891	5.7069	11.1986	1.9818	10.3685	20.1215	4.9279	23.196	47.0688	11.3029
NBE2	0.54	2.681	5.0186	0.9602	4.4708	8.7204	1.5203	7.8738	15.058	3.9888	18.5809	37.9638	8.9481
YC1	0.4953	0.8374	1.2131	0.4835	0.9351	1.485	0.4778	1.0751	1.9351	0.5366	1.5134	2.6161	1.1336
YC2	0.4029	1.5708	2.8655	0.5477	2.2782	4.3429	0.7051	3.2017	6.5876	1.4717	6.6805	13.5752	3.6858
MAULE	0.9148	0.9308	0.9592	0.851	0.8607	0.9091	0.8785	0.8117	0.9404	0.6807	2.163	1.3122	1.0177
MAUTP	0.2993	1.4027	2.6406	0.4489	2.0147	3.9568	0.6152	3.1967	6.3722	1.5841	8.1408	17.4297	4.0085
MRT	0.4963	2.4188	4.6359	0.8328	3.9734	7.6899	1.2893	6.7488	13.0209	3.3464	15.9675	32.3497	7.7308
DK	0.961	4.9275	9.5592	1.7783	8.669	17.0144	3.2171	16.5891	32.5165	13.5108	66.1047	132.8277	25.6396
GTP	0.6639	2.8529	5.3036	1.0926	4.7591	9.2655	1.7091	8.484	15.9623	4.6315	21.4102	42.9287	9.922
MLRT	0.9644	4.9726	9.6848	1.6088	8.3593	16.4964	2.1225	10.5402	21.4347	0.3111	0.5634	0.8716	6.4942
NBR	1.0919	1.0278	1.0162	1.1127	1.0395	1.0252	1.1252	1.054	1.0449	1.1821	1.2572	1.3818	1.1132
BTP	0.2506	1.1588	2.262	0.5451	2.4826	5.0442	1.0546	5.0296	9.9273	2.3363	10.1268	20.5548	5.0644
NTP	1.5471	7.8549	15.4848	2.9404	14.5789	28.6603	5.4136	27.7419	54.6015	22.4348	110.9206	222.2359	42.8679
NRT	0.9989	5.0537	9.8277	1.7455	8.6192	16.8208	2.9028	15.2061	29.7161	8.8394	42.5068	86.6908	19.0773
LKL	0.8869	4.538	8.8091	1.5084	7.368	14.4805	2.3632	12.1538	24.0509	5.1148	25.333	50.8274	13.1195
TPR	0.0378	0.2946	1.1689	0.0333	0.3205	1.5699	0.0322	0.5864	4.0627	0.0602	4.453	30.9083	3.6273
NTPR	0.0114	0.0871	0.26	0.0086	0.1856	0.0165	0.0109	0.1745	0.3418	0.0512	0.0804	0.0961	0.1103

Table 2. Cont.

	(3)												
ρ	0.8	0.8	0.8	0.9	0.9	0.9	0.95	0.95	0.95	0.99	0.99	0.99	
σ	1	5	10	1	5	10	1	5	10	1	5	10	Average
OLS	5.4148	26.507	53.7244	11.0543	55.6874	112.0445	23.0374	116.724	230.1086	130.3802	635.3272	1278.3868	223.1997
LTE	2.0315	10.251	20.8053	2.4438	12.7174	25.9111	2.4122	12.2627	24.3903	2.6611	12.1717	24.5565	12.7179
TP1	5.049	24.9894	50.7581	9.9328	50.8289	102.741	19.7745	101.3039	199.7072	88.9972	435.58	876.7175	163.865
TP2	1.6752	8.9828	18.4236	2.8205	14.9789	31.2256	4.79	24.6148	47.701	13.1524	62.4941	127.6985	29.8798
NBE1	1.5576	8.2418	16.8023	2.9397	15.5306	32.1604	5.0261	26.1854	51.4893	9.6223	46.4428	94.9991	25.9165
NBE2	0.8805	4.6735	9.5612	1.5039	7.7953	16.3748	2.4562	12.5879	24.4066	5.631	26.9415	55.7496	14.0468
YC1	0.5596	0.6861	0.9016	0.4932	0.6787	0.9801	0.4616	0.7575	1.1479	0.4482	0.9098	1.5479	0.7977
YC2	0.3781	1.6272	3.2751	0.5036	2.3652	4.7322	0.7067	3.3929	6.5462	1.6449	7.6278	15.4208	4.0184
MAULE	0.9366	0.9366	0.9552	0.8805	1.1261	0.8788	0.8044	0.8982	1.0985	0.8392	0.7937	3.2307	1.1149
MAUTP	0.3586	1.9267	3.9458	0.559	2.963	6.0918	0.8753	4.5595	8.9804	2.3608	11.4885	24.6318	5.7284
MRT	0.718	3.9049	7.9827	1.1975	6.4897	13.3097	2.0246	10.5282	20.8656	5.3812	26.2046	53.9118	12.7099
DK	2.3527	12.0831	24.6936	4.6704	23.8356	48.7979	9.3451	47.293	92.7001	45.1835	220.0866	441.4248	81.0389
GTP	0.9909	4.6104	9.3906	1.581	7.9511	16.5769	2.7194	13.3108	25.5013	7.175	33.5146	67.9357	15.9381
MLRT	2.4807	12.7349	26.1541	4.1755	21.3299	42.0605	4.388	20.7446	39.9394	0.4021	0.4534	0.538	14.6168
NBR	1.0724	1.0198	1.0107	1.095	1.0275	1.0152	1.1014	1.0338	1.0193	1.0872	1.0636	1.0871	1.0528
BTP	0.6555	3.122	6.2425	1.5083	7.2631	14.7341	2.7218	12.6039	24.9629	4.4999	18.4125	35.4053	11.011
NTP	4.0038	19.9154	40.5459	7.5735	38.469	77.9173	14.5782	73.8364	145.117	67.0577	327.0009	656.2634	122.6899
NRT	2.1217	11.2662	23.1603	3.7177	20.002	41.5298	6.6279	34.6753	68.4198	21.0663	103.2596	209.4695	45.443
LKL	2.1426	11.0032	22.5011	3.9113	19.9754	40.7736	6.8193	34.6914	68.4115	17.2726	84.7935	170.1028	40.1999
TPR	0.0257	0.1487	0.723	0.0191	0.1709	0.9609	0.019	0.2726	1.9985	0.04	2.2514	20.9858	2.3013
NTPR	0.0071	0.0258	0.2659	0.0367	0.0485	0.1531	0.0264	0.1079	0.0429	0.0009	0.5079	0.0027	0.1022

The results in Table 2(1)–(3) reveal substantial deterioration of OLS and classical estimators, whose MSEs rose by factors of 20–50. However, NTPR remained remarkably stable. Its average MSE increased only marginally compared to OLS in Table 2(1). This confirms that the adaptive eigenvalue-based shrinkage in NTPR provides implicit robustness, even without explicit outlier weighting. From the practical standpoint, the simulation results suggest that the proposed NTPR estimator is particularly effective when the correlation among predictors exceeds 0.90 and the sample size is small to moderate. In such scenarios, classical ridge and Liu-type estimators may still suffer from variance inflation, whereas NTPR provides more stable estimation with substantially reduced MSE.

4.3. Real-Life Application

To demonstrate the practical example of the proposed estimator, an empirical analysis is conducted using the “Body Fat” dataset, which was originally published in the textbook, *Biostatistics for the Biological and Health Sciences*, second edition, by Triola et al. [30], and is publicly available at www.triostat.com/biostat3e accessed on 21 June 2025. The dataset consists of measurements collected from 300 adult subjects and has been widely used in statistical modeling and regression analysis, due to the presence of strong correlations among variables.

For the analysis, the following model is considered:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_4 X_4 + \epsilon,$$

where the response variable Y denotes a high-density lipoprotein (HDL) cholesterol level (mg/dL), an important biomarker associated with cardiovascular health. The explanatory variables include body weight (X_1 , in kg), height (X_2 , in cm), waist circumference (X_3 ,

in cm), and arm circumference (X_4 , in cm). These anthropometric measures are biologically related and therefore expected to exhibit substantial multicollinearity.

Preliminary diagnostic measures, including pairwise correlations, the variance inflation factor (VIF) and the condition number of the design matrix, confirmed the presence of moderate to severe multicollinearity, making this dataset well suited for evaluating the performance of biased regression estimators that are designed to address multicollinearity.

The evidence of strong substantial multicollinearity among the predictors is apparent from the correlation matrix in Table 3 and the fact that one of the variance inflation factors (VIFs) exceeds 10 (see Ozkale and Kaciranlar 2007) [8], and it is considered to be the threshold for multicollinearity (Table 3). Additionally, the structure of the correlation matrix resembles a block correlation pattern (Table 3).

Table 3. Correlation matrix.

	y	x ₁	x ₂	x ₃	x ₄
y	1				
x ₁	−0.3168	1			
x ₂	−0.2035	0.3637	1		
x ₃	−0.3197	0.8945	0.0778	1	1
x ₄	−0.2971	0.8880	0.1768	0.8024	1

We can see that the independent variables in this dataset also exhibit high levels of correlation. Also, we can use the variance inflation factor (VIF) to check the multicollinearity, which is calculated as follows:

$$VIF_i = \frac{1}{1 - R_i^2}, i = 1, \dots, p,$$

where R_i^2 denotes the multiple correlation coefficient, which is obtained from regression of the explanatory variable, and the VIF values are as follows:

Based on the VIF values from Table 4, variables such as x_1 (17.04) and x_3 (8.11) exhibit high multicollinearity, so we can conclude that there is a dependency among these explanatory variables.

Table 4. Variance inflation factor.

Variable	VIF
x ₁	17.04
x ₂	2.11
x ₃	8.11
x ₄	5.61

To further assess the severity of the multicollinearity, we obtained the condition number value, calculated as $CN = \left(\frac{\text{largest eigen value}}{\text{smallest eigenvalue}} \right)^{1/2} = 88.41461$. The resulting condition number is 88.41461, which exceeds the commonly used threshold of 10, confirming the presence of severe multicollinearity (Liu, 2003 [7]) among the variables.

We have checked the assumption of multicollinearity for the OLS, and we have found that they have severe multicollinearity among the independent variables. We have also checked the assumptions for the homoscedasticity and independence of errors, and from Figure 2, we have found that the residuals versus fitted values plot shows a random scatter of residuals around zero with no systematic pattern or change in the spread, indicating

that the assumption of homoscedasticity is satisfied. Additionally, the residuals display no discernible trend or clustering, suggesting that the errors are independent.

In this study, the theoretical conditions are established in Theorems 1–5, using the dataset. The outcomes of this evaluation are reported in Table 5, which shows that all condition values for the dataset are less than one, which is consistent with the requirements specified in the theorems.

Table 6 reports the estimated regression coefficients and also the mean squared error (MSE) values applied to the dataset. The results clearly indicate that all estimators produce smaller MSE values compared to the ordinary least squares (OLS) estimator, although the magnitude of improvement varies across methods. Some estimators yield performance levels that are relatively close to OLS, while others demonstrate substantial gains in estimation accuracy under multicollinearity.

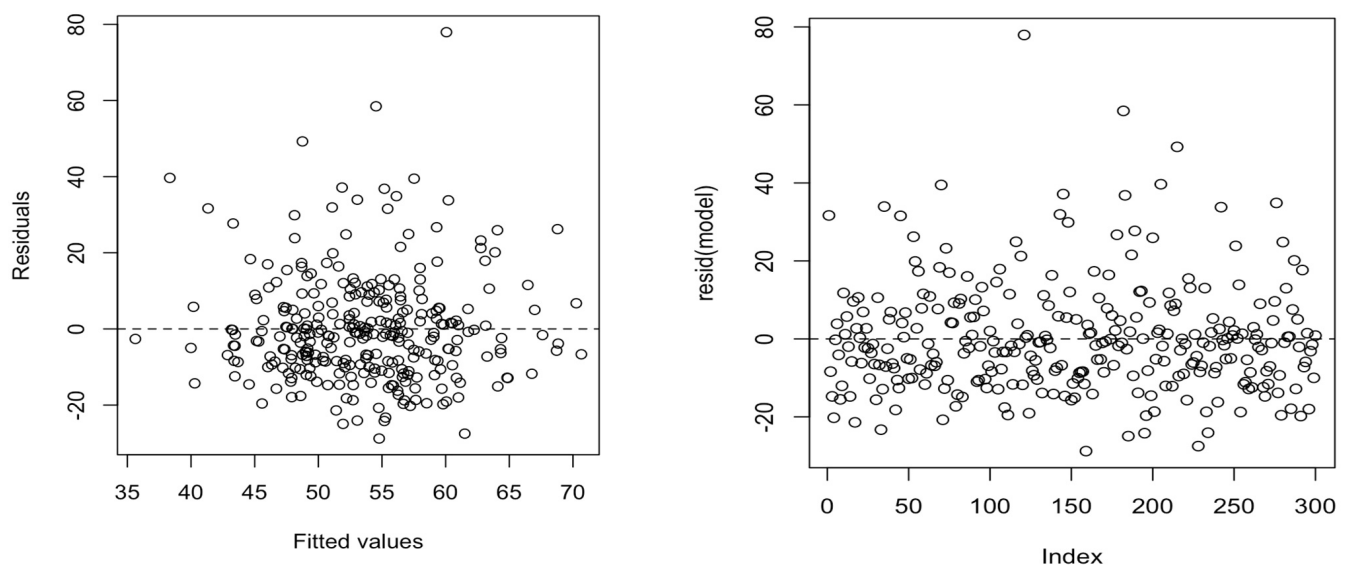


Figure 2. Residual-based diagnostic plots for evaluating regression assumptions.

Table 5. Checking the validation of the theoretical conditions for the real-life data.

Theorems	Conditions	Value
2.1	$b_{NTPR}^T \left[\sigma^2 \left(\Lambda^{-1} - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) \right]^{-1} b_{NTPR} < 1$	0.01142875
2.2	$b_{NTPR}^T \left[\sigma^2 \left(A_{LTE} \Lambda^{-1} A_{LTE}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) + b_{LTE} b_{LTE}^T \right]^{-1} b_{NTPR} < 1$	0.0179153
2.3	$b_{NTPR}^T \left[\sigma^2 \left(A_{TP} \Lambda^{-1} A_{TP}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) + b_{TP} b_{TP}^T \right]^{-1} b_{NTPR} < 1$	0.0103903
2.4	$b_{NTPR}^T \left[\sigma^2 \left(A_{NBE} \Lambda^{-1} A_{NBE}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) + b_{NBE} b_{NBE}^T \right]^{-1} b_{NTPR} < 1$	0.01037587
2.5	$b_{NTPR}^T \left[\sigma^2 \left(Y_{k,d} \Lambda^{-1} Y_{k,d}^T - A_{NTPR} \Lambda^{-1} A_{NTPR}^T \right) + b_{YC} b_{YC}^T \right]^{-1} b_{NTPR} < 1$	0.01197182

We applied all estimators from Section 2 to this dataset. Table 5 summarizes the theoretical condition checks from Theorems 1–5; all values are less than one, confirming that the assumptions required for MSE dominance of the NTPR estimator over the competing estimators are satisfied.

The estimated regression coefficients and corresponding MSEs are provided in Table 6. While the OLS estimator produced MSE = 0.1545, most biased estimators achieved smaller errors: LTE (0.1314), TP2 (0.1146), and MRT (0.0771). Notably, the proposed NTPR estimator

attained the lowest MSE ($\approx 1.7 \times 10^{-5}$): a reduction of nearly 99.99%, relative to OLS. This dramatic improvement indicates that the eigenvalue-adaptive shrinkage mechanism of NTPR effectively stabilizes the estimates without introducing excessive bias.

Table 6. Regression analysis and MSE of the data.

	OLS	LTE	TP1	TP2	NBE1	NBE2	YC1	YC2	MAUTP	MRT
X_1	−0.5773	−0.4961	−0.5457	−0.5307	−0.5479	−0.5454	−0.5414	−0.5427	−0.5081	−0.5178
X_2	0.3378	0.3317	0.3481	0.3456	0.348	0.3481	0.3482	0.3482	0.3611	0.357
X_3	0.1891	0.1687	0.192	0.1749	0.194	0.1917	0.1881	0.1893	0.1983	0.1939
X_4	0.7588	0.6466	0.6197	0.6458	0.6198	0.6197	0.62	0.6199	0.4414	0.4995
MSE	0.1545	0.1314	0.1092	0.1146	0.1298	0.1297	0.1088	0.1089	0.0643	0.0771
	DK	GTP	MLRT	NRT	LKL	TPR	NTPR			
X_1	−0.5485	−0.4484	−0.5492	−0.5477	−0.5701	0.0928318	0.0928147			
X_2	0.3478	0.3619	0.3478	0.3479	0.3405	0.1946531	0.1946173			
X_3	0.1941	0.1338	0.1948	0.1934	0.191	0.1135197	0.1134988			
X_4	0.6216	0.4824	0.6216	0.6217	0.7217	0.0379703	0.0379633			
MSE	0.1101	0.1082	0.1102	0.11	0.1417	0.0000174	0.0000173			

Coefficient patterns further reinforce this conclusion. While OLS yielded unstable magnitudes, NTPR produced smoother, moderate coefficients with consistent signs and reduced variability. These results align with the theoretical expectations from Section 3.1, showing that NTPR simultaneously controls variance inflation and numerical instability.

For practitioners, these findings indicate that NTPR can serve as a reliable alternative to traditional biased estimators when multicollinearity is severe and coefficient instability is a concern. The dramatic reduction in MSE demonstrates that the estimator is not only theoretically superior but also practically useful in real-world regression analysis.

5. Conclusions and Directions for Future Research

This study proposed a new minimum eigenvalue two-parameter ridge (NTPR) estimator for the Gaussian linear regression model to mitigate the challenges of multicollinearity. The proposed estimator was examined alongside a range of two-parameter estimators, including the ordinary least squares (OLS), Liu type estimator, modified ridge-type estimator, unbiased two-parameter estimator, and yang and Chang estimator, to name some. For each estimator, we derived the expressions of the bias, covariance, and also the mean squared error (MSE), all of which depend on the biasing parameters k and d . Theoretical comparisons were conducted, using MSE as the primary criterion.

A unified theoretical framework was established to build a performance comparison between the proposed NTPR estimator and 16 existing two-parameter estimators under the MSE framework. Also, a comprehensive Monte Carlo simulation study has been conducted to evaluate estimator behavior under various scenarios. The simulation design incorporated different sample sizes, number of predictors, standard errors and degrees of multicollinearity among the regressors. Each scenario involved 5000 replications, and the average MSE values were calculated to assess estimator performance.

The simulation results indicated that higher multicollinearity among predictors generally resulted in increased MSE values. Among all estimators, NTPR, and TPR outperformed the others. Estimator performance improved with the increasing sample size, as reflected by the declining MSE values, as the increasing number of predictors led to lower MSE. NTPR achieved the smallest MSE and produced stable, interpretable coefficients, even

under 25% outlier contamination. These findings suggest that the NTPR estimator performs favorably, especially when compared to the OLS estimator, which consistently yielded higher MSE values. The practical applicability of the proposed methods is illustrated using body fat data.

Although the proposed estimator demonstrates strong theoretical properties and superior empirical performance, several limitations of the present study should be acknowledged. First, the theoretical dominance results are derived under the classical linear regression framework with normally distributed errors and a fixed design matrix. While this assumption is standard in the literature, deviations from normality may affect the estimator's performance in practice.

Second, the simulation study focuses on low to moderate dimensional settings with a limited number of predictors. Although this setting is relevant for many applied problems, the behavior of the proposed estimator in high-dimensional scenarios, where the number of predictors may exceed the sample size, remains an open question.

Third, the tuning parameters of the proposed estimator are selected based on analytical considerations and empirical performance. Alternative data-driven selection procedures, such as cross-validation or information-criterion-based approaches, may further enhance its practical implementation.

These limitations naturally motivate several directions for future research. Extending the proposed estimator to generalized linear models and high-dimensional regression frameworks is a promising avenue for further study. In addition, developing robust versions of the estimator to accommodate heavy-tailed error distributions and outlier contamination would improve its applicability in real-world data analysis. Finally, investigating adaptive and fully data-driven tuning parameter selection methods represents an important topic for future methodological development.

The conclusions drawn from this study are specific to the simulation settings, particularly the selected combinations of sample size n , number of predictors p , standard error σ and correlation coefficient ρ . Although several methods exist in the literature for estimating the biasing parameters k and d , only a subset was considered in this work. Future work may extend this framework by exploring other generalized linear models, like Logistic, Poisson, Beta, or Negative binomial; finding type I error values and power properties; and also exploring the robust weighted variants of NTPR, thereby broadening its applicability in high-dimensional and contaminated data environments.

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